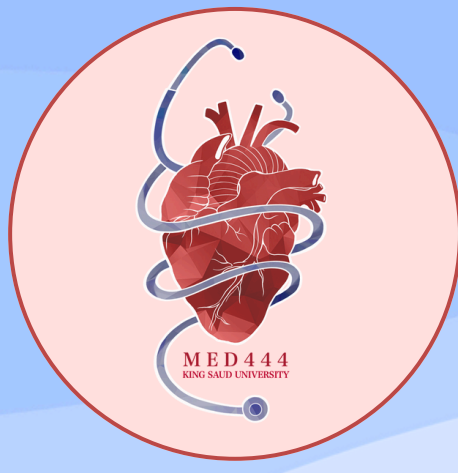




AI in healthcare

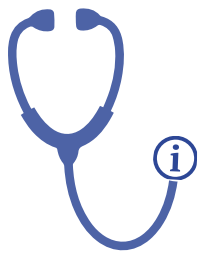
Editing files

- main text- male slides
- female slides- important
- dr notes - textbook
- extra / old slides

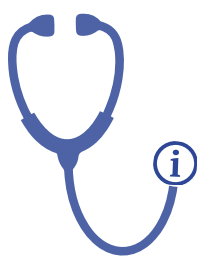
objective



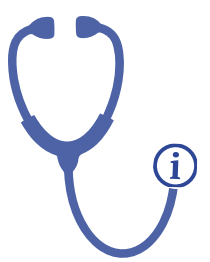
Define the Role of AI in Healthcare



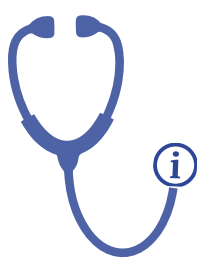
Identify Key AI Concepts



Recognize AI Applications in Diagnosis and Treatment



Discuss Enhanced Patient Care with AI



Analyze Ethical and Regulatory Aspects



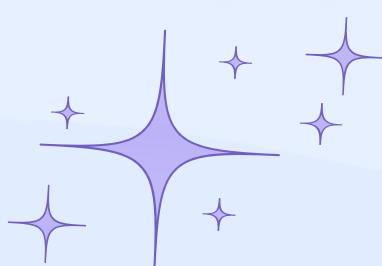
Anticipate Future AI Trends in Healthcare



Analyze Case Studies and Real-World Examples

IMPORTANT!

The Stars in this lecture are the summary of Dr. Rania
She said in the recording: "These are the important
points I want you to focus on."



اللهم انفعني بما علّمتني، وعلمني ما ينفعني، وزدني علماً



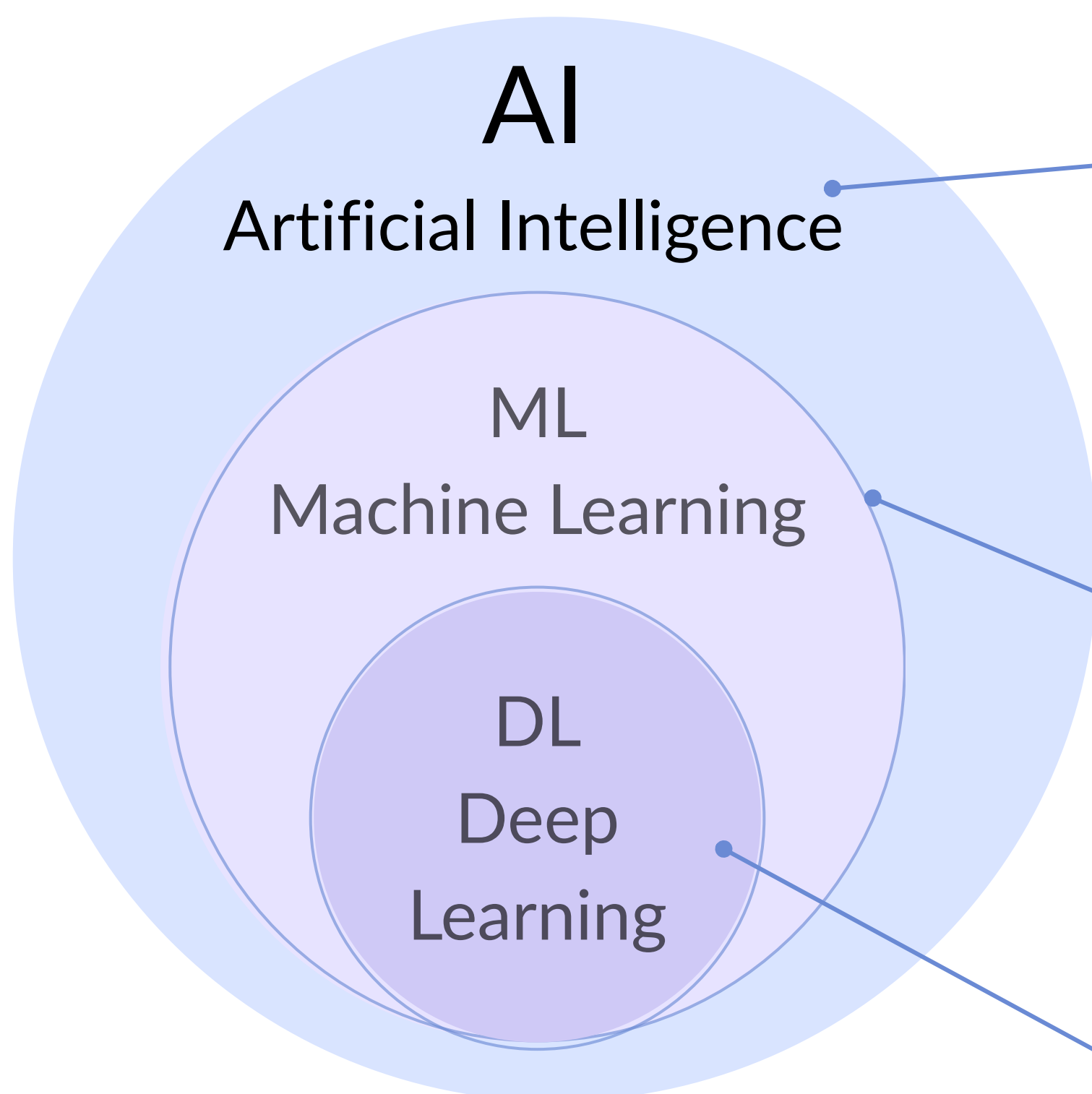
Whole Page

DEFINE THE ROLE OF AI IN HEALTHCARE



What is AI?

- Computer systems that recognize patterns.
- Multidisciplinary field of study and research involve computer system and algorithm and capable of performing task that requires human intelligence.
- Machine or system associated with HI.



- stimulation of human intelligence and beyond using programmatic routines.

- Broadest one include ML and DL
- Mimic human cognitive function

- Subset of AI.
- Uses statistical algorithms to represent complex patterns within data.
- Development algorithm that allow to computer to learn from data, make decisions based on this data

- Subset of ML.
- Uses layered artificial neural networks capable of emergently representing latent properties of the data.
- Inspired by structure of human brain.
- successful in task involved unstructured data handle large and complex data and task

Why use AI in healthcare?

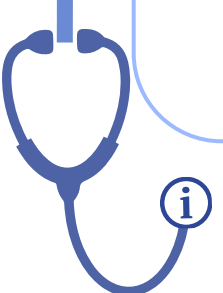
- Improve patient outcomes.
- automate processes and save time.
- Reduce costs.

Why use AI in Mayo Clinic?

- Administrative Efficiency.
- Predictive analytics.
- Diagnosis and disease detection.
- Drug and discovery development.

What does AI lack?

- Creativity.
- Empathy.
- Compassion.





Identify Key AI Concepts



Machine learning is not magic

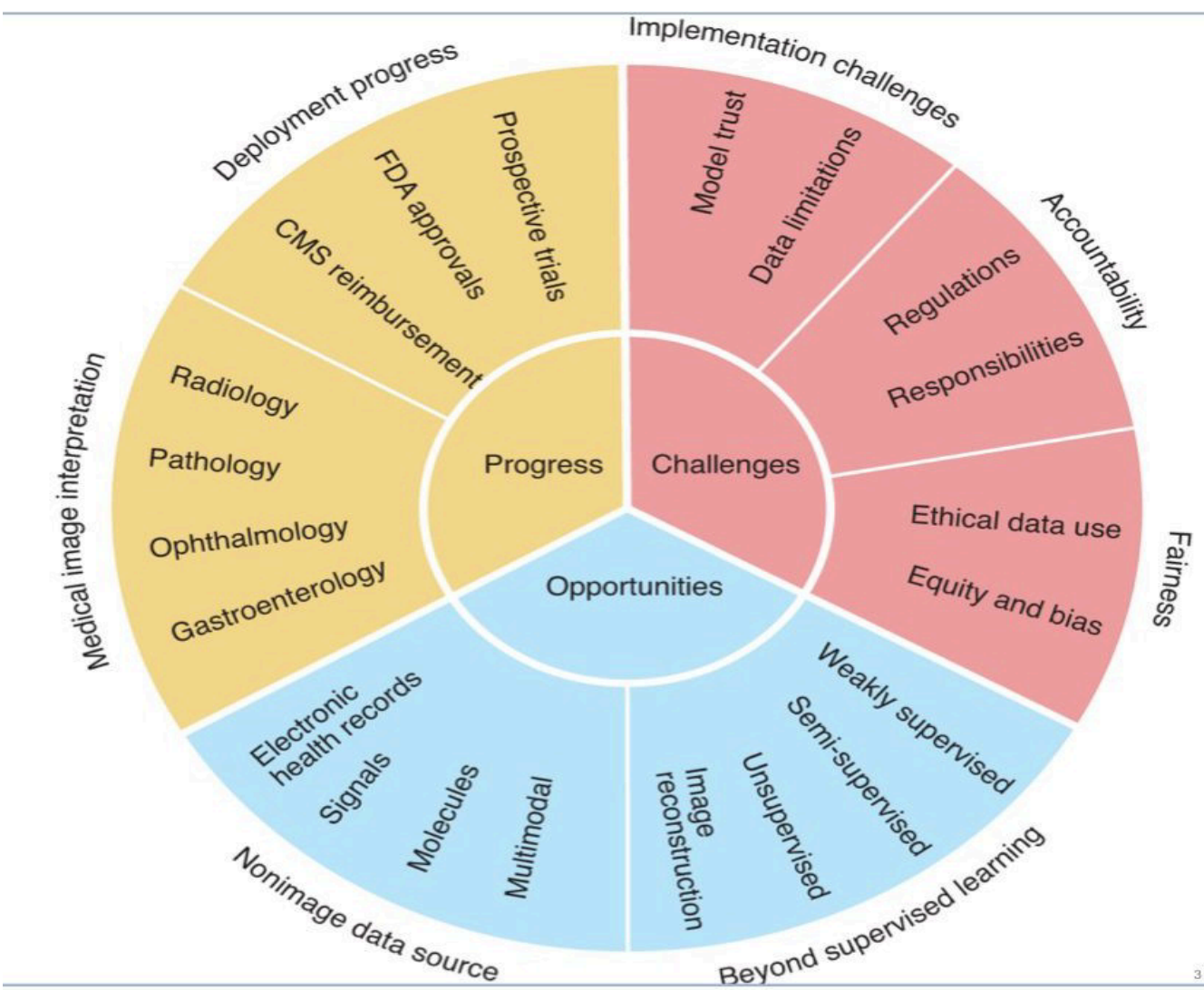
- It learns pattern; it doesn't explain causation.
- It cannot replace clinical experience, domain knowledge, clinical decision.
- It cannot replace clinical intuition and qualitative assessment.
- It cannot handle rare, unpredictable events.
- Who can do all these things? Human.

AI key concepts

Self-supervised learning	Learning from unlabeled data by leveraging information extracted from the data itself
Semi-supervised learning	Learning from a small amount of labeled data combined with a large amount of unlabeled data
Causal inference	Finding the effect of a component or treatment on a system using data
Reinforcement learning	Learning in an interactive environment using feedback from actions and past experiences

- AI setup beyond supervised learning.

Overview of the progress, challenges and opportunities for AI in health



Recognize AI Applications in Diagnosis and Treatment



Examples of applications of AI in healthcare

- Radiology (medical image)
- Pharmacy
- Cardiology Research
- EHR
- Scheduling
- Sepsis (diagnosis)
- Surgery
- Text
- Strategy
- Speech
- Language translation
- Readmission
- Risk
- Chatbots
- Wearable devices
- Socio-demographic
- Dermatology
- Digital assistant
- Screening
- Nuclear medicine
- Face recognition
- Emergency medicine
- Triage
- Faster diagnosis
- Lab tests
- Robotics
- Radiation therapy
- Education
- Genetic
- Pathology and
- Histology (analyze slides)

Examples of problem and challenges to solves with AI

- Improve diagnosis
- Faster diagnosis
- Reduce errors
- Information retrieval
- Improve patient safety
- Improve treatment
- Faster treatment
- Reduce waiting time
- Self-service and self-care
- Optimize resource usage
- Scheduling and logistics
- Risk stratification
- Better decisions
- Patient privacy
- Communication
- Minimize administrative work
- Improve research
- Improve efficiency
- Reduce bias
- Monitoring
- Expand access
- Patient preparation

Discuss Enhanced Patient Care with AI

1

Cardiology (patterns in 12-lead ECGs)

- Low ejection fraction
- Atrial fibrillation
- Hypertrophic cardiomyopathy
- Cardiac amyloidosis
- Aortic valve stenosis

Can be done with BNP

An artificial intelligence-enabled ECG algorithm for the identification of patients with atrial fibrillation during sinus rhythm: a retrospective analysis of outcome prediction

Zachi I Attie¹, Peter A Noseworthy², Francisco Lopez-Jimenez³, Samuel J Asirvatham⁴, Abhishek J Deshmukh⁵, Bernard J Gersh⁶, Rickey E Carter⁷, Xiaoxi Yao⁸, Alejandro A Rabinstein⁹, Brad J Erickson¹⁰, Suraj Kapa¹¹, Paul A Friedman¹²

Summary

Background Atrial fibrillation is frequently asymptomatic and thus underdetected but is associated with stroke, heart failure, and death. Existing screening methods require prolonged monitoring and are limited by cost and low yield. We aimed to develop a rapid, inexpensive, point-of-care means of identifying patients with atrial fibrillation using machine learning.



Lancet 2019; 394: 861-67
Published Online
August 1, 2019
[http://dx.doi.org/10.1016/S0140-6736\(19\)31721-0](http://dx.doi.org/10.1016/S0140-6736(19)31721-0)

2

Computational Pathology (Patterns in images)

- Highlight regions of interest in a slide
- Detect presence of cancer
- Tumor grading
- Identify similar cases
- Predict mutation status

Impact of Artificial Intelligence on Miss Rate of Colorectal Neoplasia

Michael B Wallace¹, Pradeep Sharma², Pradeep Bhandari³, James East⁴, Giulio Antonelli⁵, Roberto Lorenzetti⁶, Michael Veth⁷, Maria Speranza⁸, Marco Spadaccini⁹, Mathias Desai¹⁰, Frank J Lukens¹¹, Daniel Babiarz¹², Casey Batista¹³, Desvinder Singh¹⁴, William Palmer¹⁵, Francisco Ramirez¹⁶, Rebecca Palmer¹⁷, Tisha Lunford¹⁸, Kevin Huff¹⁹, Elizabeth Bird-Liebermann²⁰, Victor Ciofalo²¹, Sophie Arnold²², David Cargoni²³, Kristy Puskas²⁴, Gregory Clarke²⁵, Ambarish Juhel²⁶, Mohammed Barakat²⁷, Luigi Longo²⁸, Luigi Moro²⁹, Alessandro Ripici³⁰, Cesare Hassan³¹

	AI first (n = 116)	Standard colonoscopy first (n=114)	P value	OR [95% CI]
Subjects with ≥ 1 colorectal polyp in the 2 nd colonoscopy, n (%)	33 (28.45)	55 (48.25)	.002	0.43 [0.25-0.74]
Subjects with ≥ 1 adenoma or carcinoma in the 2 nd colonoscopy, n (%)	29 (25.00)	52 (45.61)	.001	0.40 [0.23-0.70]
Subjects with ≥ 1 adenoma or carcinoma in the 2 nd colonoscopy in subjects with a negative 1 st colonoscopy, n (%)	3 (6.82%)	13 (29.55)	.006	0.17 [0.05-0.67]

Polyp and Adenoma/Carcinoma Detection Rates in the Second Colonoscopy (Per Patient Analysis): FAS Population

Discuss Enhanced Patient Care with AI



3

Paper medical records (Patterns in text) in clinical documentation and EHR



لا إله إلا أنت سبحانك إني كنت من الظالمين

- Identify text scanned images.
- Find specific medical information.
- Reduce the amount of time to review.
- Piloted across 7 practice areas.
- 10 to 20 percent time saving.

4

Clinical narratives (Pattern in text)



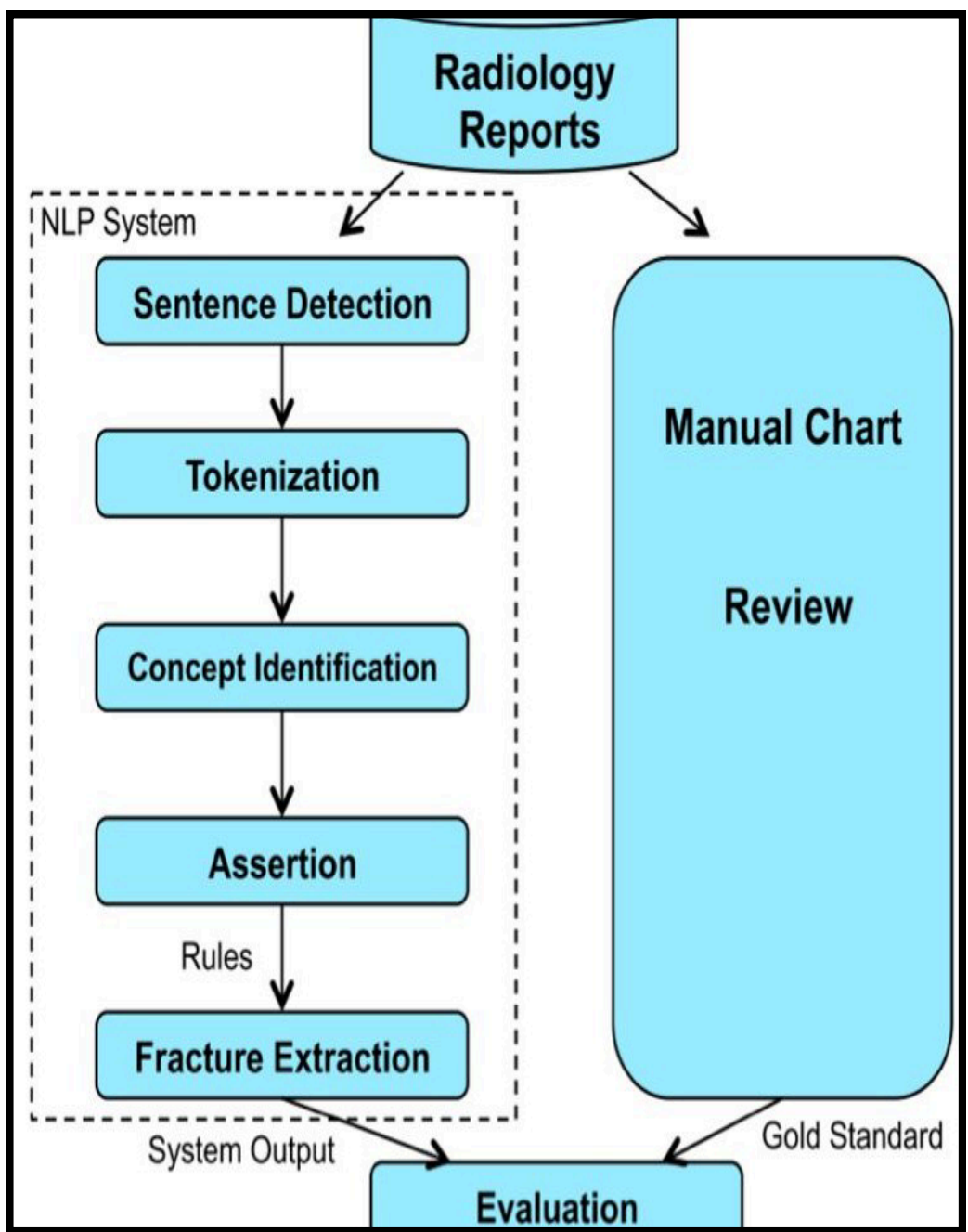
- Radiology reports, clinical history, physical exam findings, operative report.
- Detect keyword and concepts.
- Extract information.
- Appendicitis, pneumonia, potentially malignant lesions, skeletal site specific fractures, asthma status, allergy status.



5

Natural language processing (NLP)

- An NLP algorithm was developed to identify twenty specific skeletal site fractures from radiology reports.
- Empirical experiments were conducted to validate the effectiveness of the NLP algorithm using radiology reports from a community-based cohort at Mayo Clinic.
- Microaveraged results of the NLP algorithm for the twenty fractures include sensitivity (0.930), specificity (1.0), PPV (positive predictive value, 1.0), NPV (negative predictive value, 0.941), and F1-score (0.961).



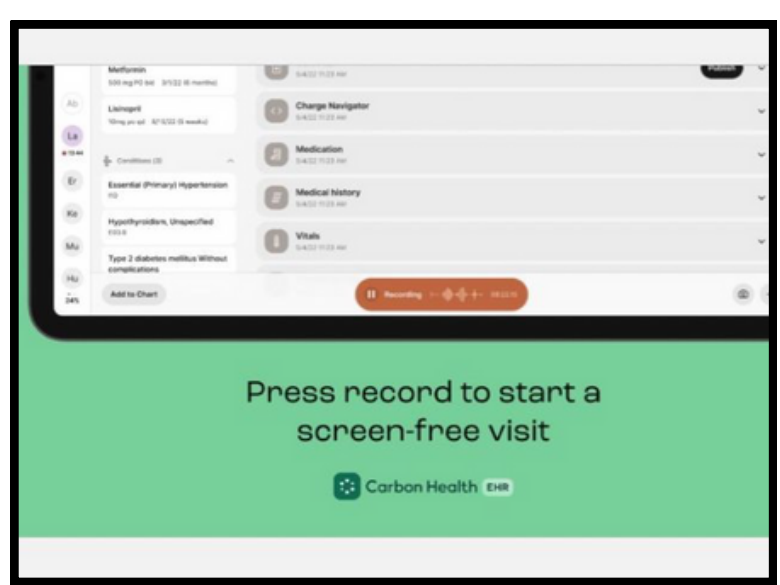
Discuss Enhanced Patient Care with AI



6

Charting (Patterns in audio and EHR data)

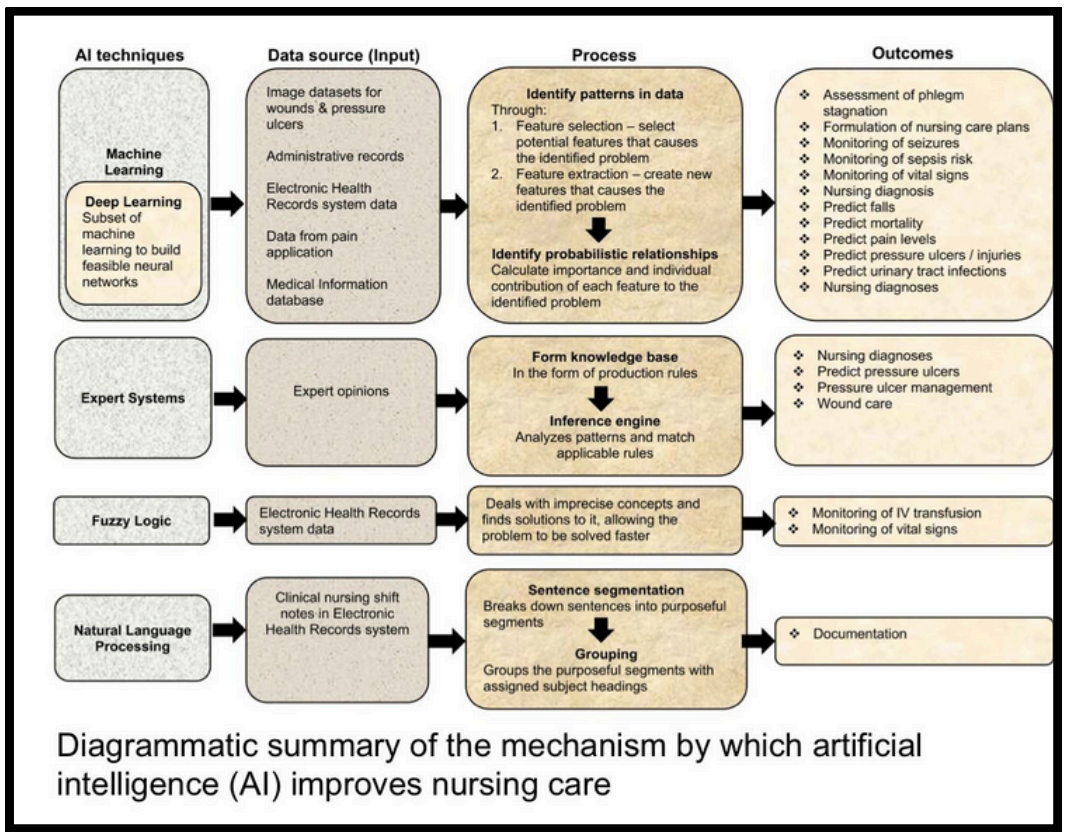
- Conversation with patient is recorded.
- Recorded audio is transcribed.
- Transcript + EHR data = LLM.
- LLM generates a visit summary.
- Clinician reviews, edits, approves.



7

Nursing care

- AI-enhanced patient monitoring systems' efficiency, particularly for tracking vital signs, reduced nurses' workload, giving them more time for non-routine and less mentally taxing tasks.
- AI systems are trained to automatically monitor patient data and detect abnormalities, especially in critical care situations.
- Timely alerts can help in managing patients effectively.
- Limited real-world implementation.



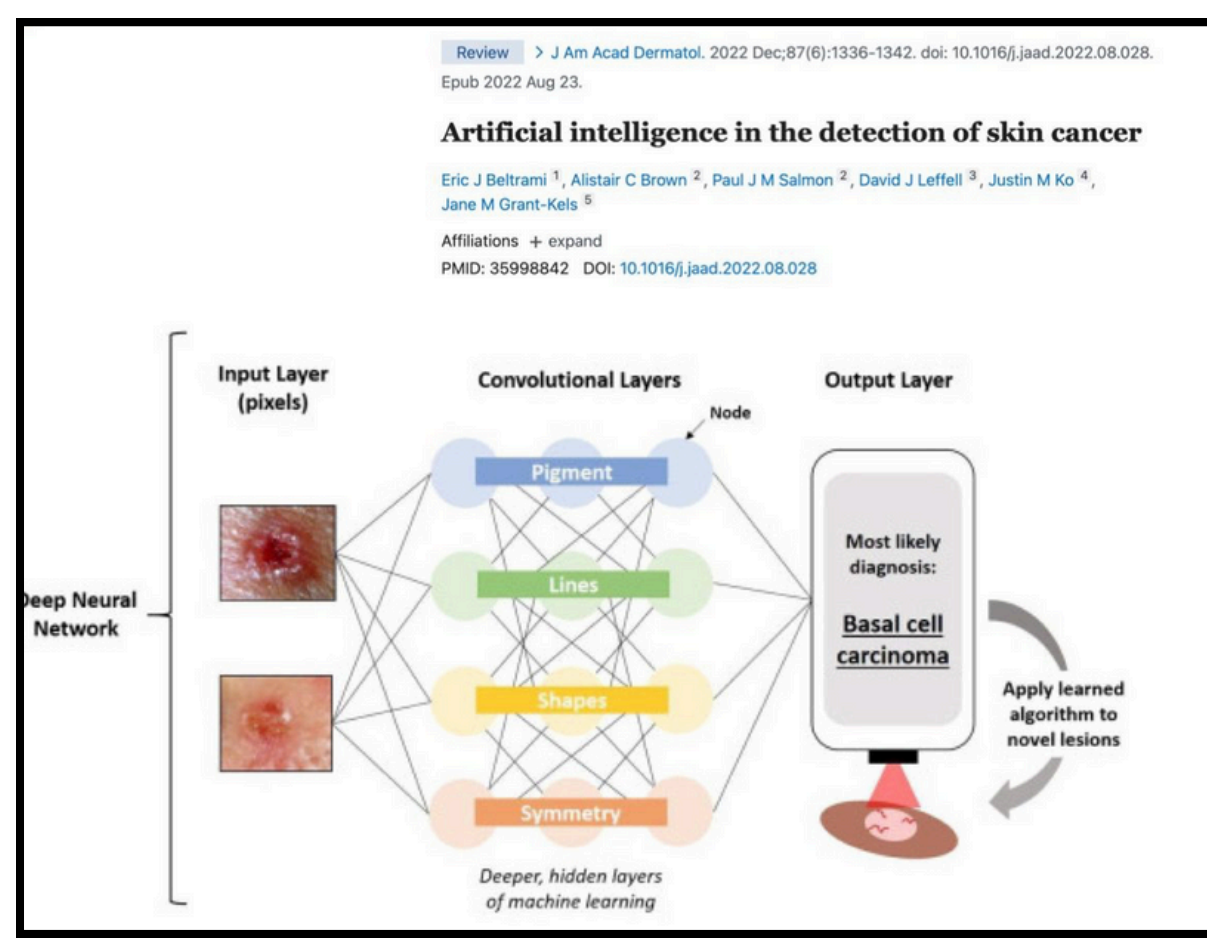
8

Dermatology



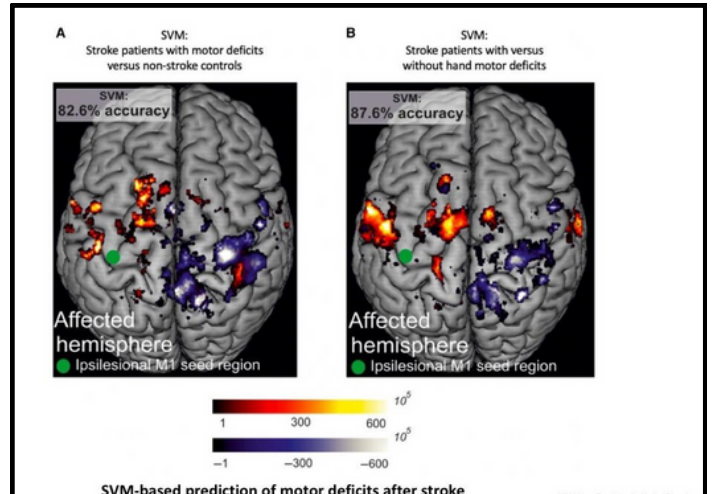
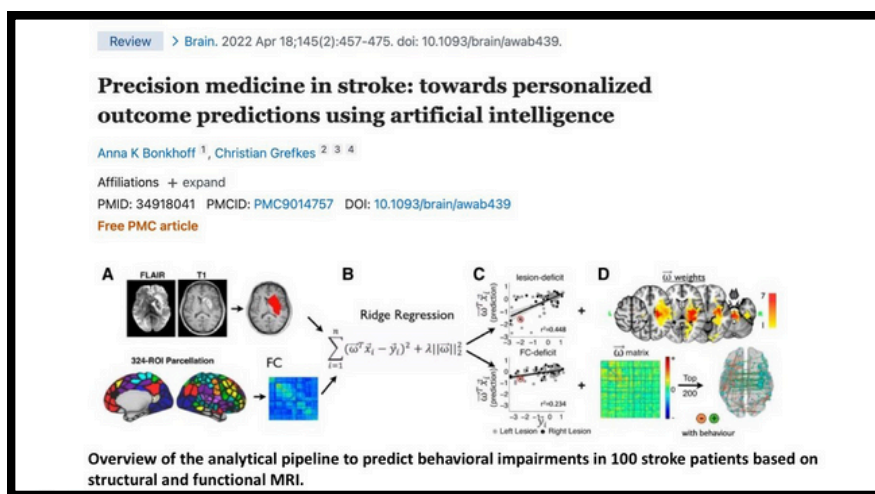
Application of deep neural networks in the identification of skin neoplasms.

- Teledermatology using AI offers dermatological expertise.
- Non dermatologist clinicians increase confidence and improve appropriateness of referrals to dermatologists.
- AI can reduce unnecessary referrals and enhance the detection of medically concerning ones, potentially expediting access to dermatologists.
- Dermatologists have a crucial role in ensuring that it is patient- centered, clinician-led, high-quality, equitable, and accessible



9

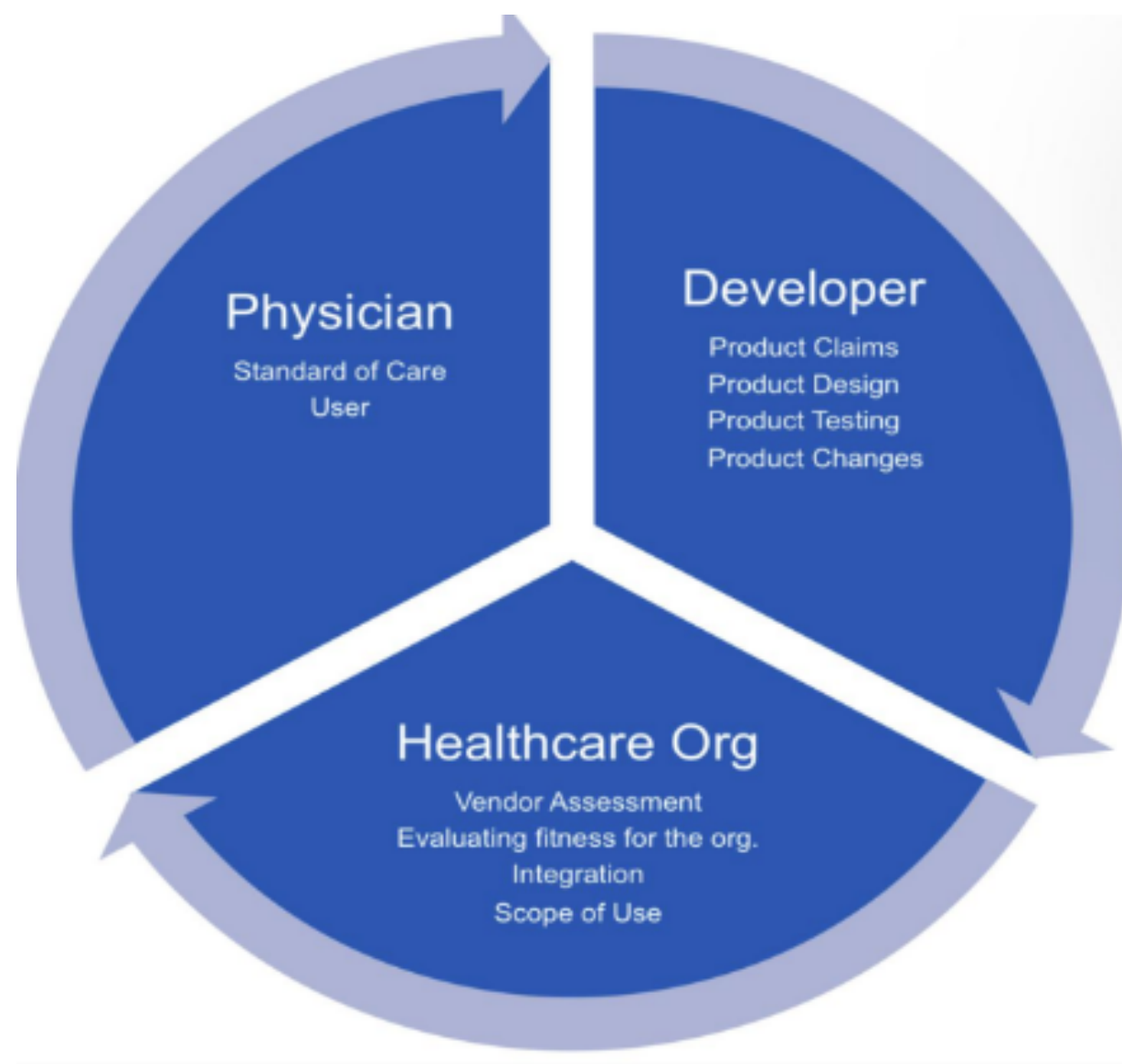
Medicine and AI



Analyze Ethical and regulatory aspects

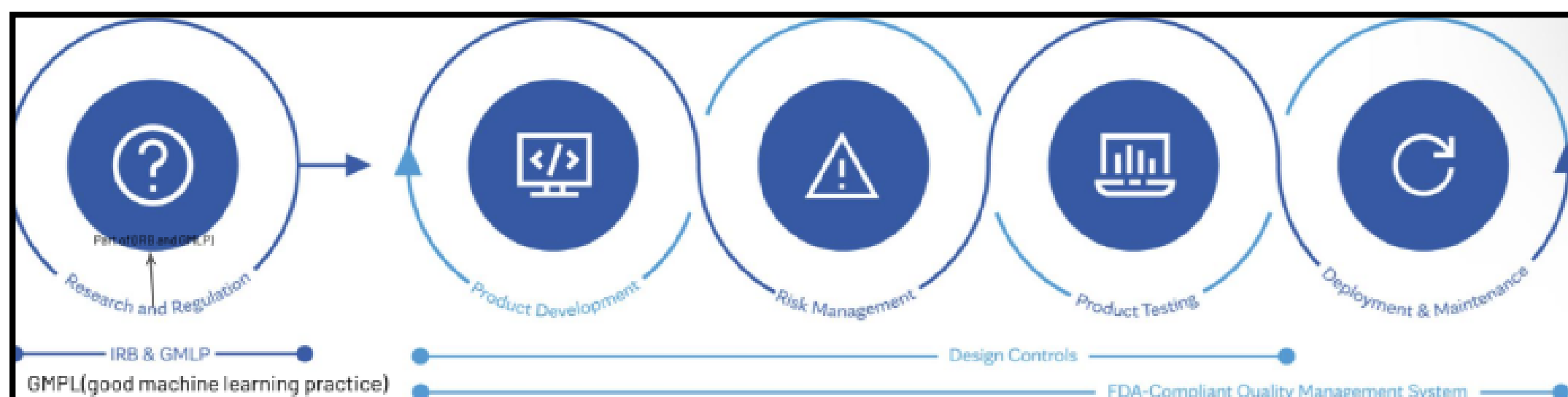


Shared accountability model examples

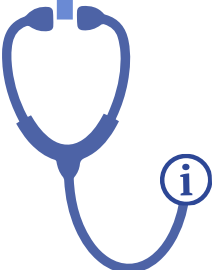
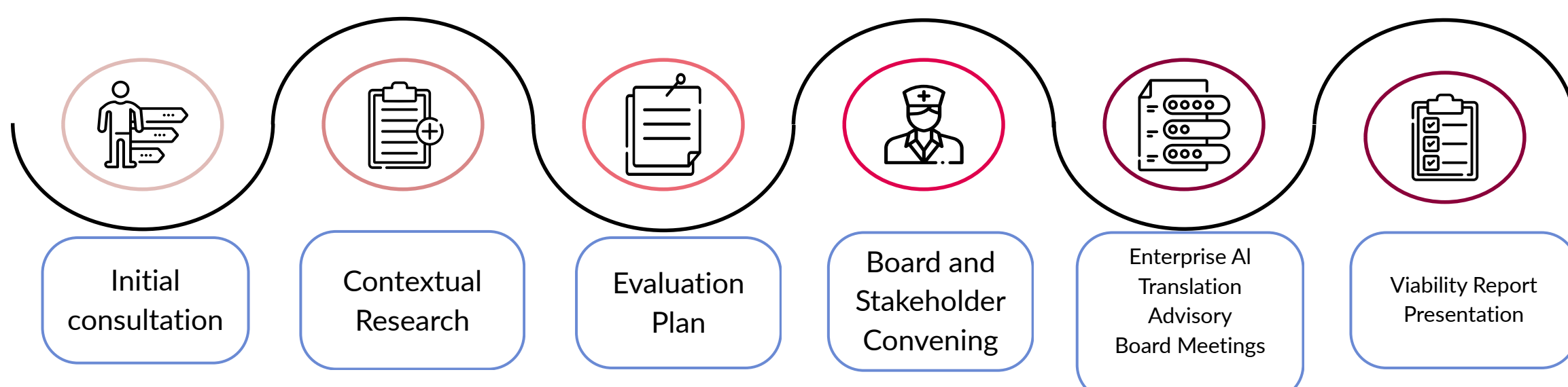


BUILDING A HEALTHCARE AI FRAMEWORK

• HUB AND SPOKE MODEL



ENTERPRISE AI TRANSLATIONAL ADVISORY BOARD

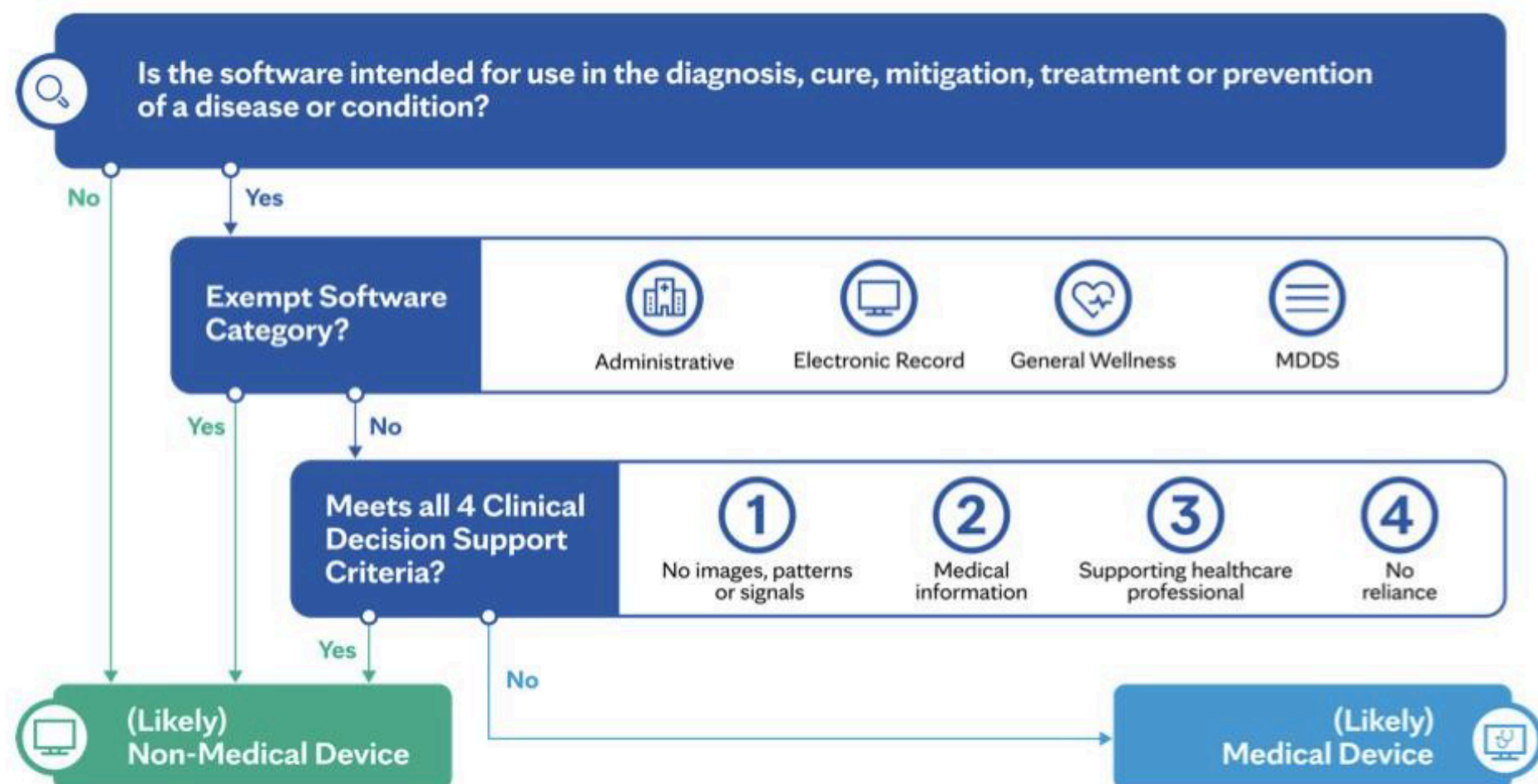




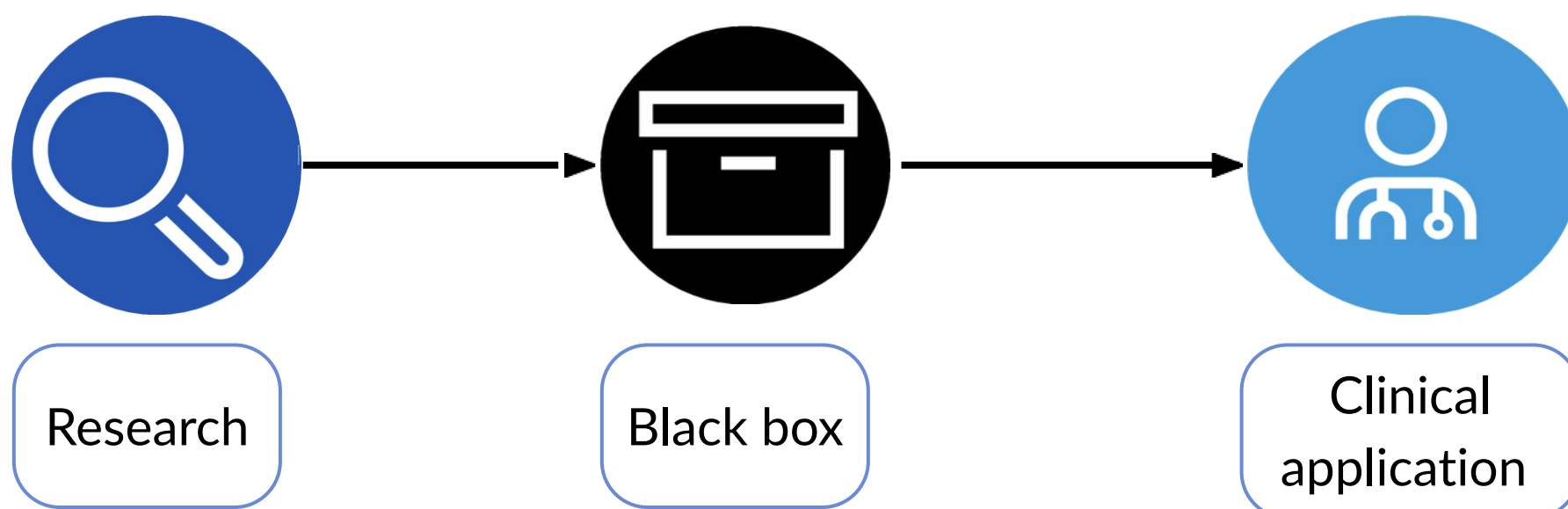
Analyze Ethical and regulatory aspects

USA STANDARDS,
IS YOUR PRODUCT A MEDICAL DEVICE?

- ANALYZING EACH PRODUCT



TRANSLATING FROM RESEARCH TO PRACTICE
• HEALTHCARE ORGANIZATIONS ENTERING
NEW TERRITORY



AI/ML RISKS

Table B.1—Examples of ML-related hazards

Performance-related hazards	Data storage/security/privacy
Data quality	Privacy failures
Incorrect data	Inability to contact patient
Incorrect handling of outliers	Overtrust
Incomplete data	Overconfidence
Subjective data	Perceived risk
Inconsistent data	User workload
Atypical data	Self-confidence
Bias	Variation in social trust
Selection bias	Use policies
Confounding variables	
Non-normality	
Proxy variables	
Implicit bias	
Group attribution bias	
Experimental bias	

TIR34971:2023

USE ERROR

Root cause related to perception

- Failure (e.o. inability) to see visual information (e.o. display is partly obscured)
- Failure (e.o. inability) to hear audio information (e.o. ambient noise)

ABNORMAL USE

- Exceptional violation (e.g. using a MEDICAL DEVICE as a hammer)
- Reckless use (e.g. using a MEDICAL DEVICE after removing its protective guards)
- Sabotage (e.g. hacking the software of a software-controlled MEDICAL DEVICE)
- Conscious disregard for the contraindications of the MEDICAL DEVICE (e.g. use on a PATIENT that has a pacemaker)

Root cause related to memory

- Memory
- Inability
- Omitting (e.g. forgetting) a planned step

IEC 62366-1: 2015

1. Conceptualization	2. Data management	3. Model development	4. Model implementation	5. Model use and decision making
Potentially unethical use cases	Incomplete or inaccurate data	Nonrepresentative data	Implementation errors	Technology-environment malfunction
Insufficient learning feedback loop	Unsecured "protected" data	Biased or discriminatory model outcomes	Poor technology-environment design	Slow detection of response to performance issues
	Other regulatory noncompliance	Model instability or performance degradation	Insufficient training and skills	Cybersecurity threats
				Failure at the human-machine interface

https://www.mckinsey.com/business-functions/quantumblack/our-insights/confronting-the-risks-of-artificial-intelligence

Evaluate and consider AI/ML risks for downstream impact

Analyze Ethical and regulatory aspects



OUR DEPLOYMENT OPTIONS

• BUILDING ON A FIRM FOUNDATION



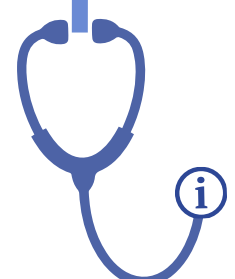
MAYO CLINIC SAMD REVIEW BOARD

• SAFE, EFFECTIVE, AND ETHICAL SOFTWARE FOR OUR PATIENTS

- Regulatory Determinations
- Risk Assessment
- Recommended Risk Controls

SOFTWARE AS A MEDICAL DEVICE

- **Regulatory Strategy:** Work with the SaMD Review Board to create a regulatory pathway.
- **Roles & Responsibilities:** Determine who needs to do what, when.
- **Development & Testing Process:** Follow the regulations to prove your medical device systems is designed, developed, and tested to be safe and effective.
- **Validation Study:** Work with the IRB to conduct a study that proves you built the right system in its intended setting,
- **Submit to the FDA:** Put all your hard work together and submit to the FDA.
- **Post Deployment Operations:** After deployment, conduct oversight, monitoring, and change management.





Analyze Ethical and regulatory aspects

CHALLENGES

- Ethical challenges for AI in medicine

Uncertainties and distrust of AI predictions

Regulation and governance of medical AI

Shifts in responsibility introduced by using AI in practice

Concerns about data privacy and security

Develop explainable algorithms and transparent model reporting

Standardize safe AI practices and establish through, transparent reporting performance reporting

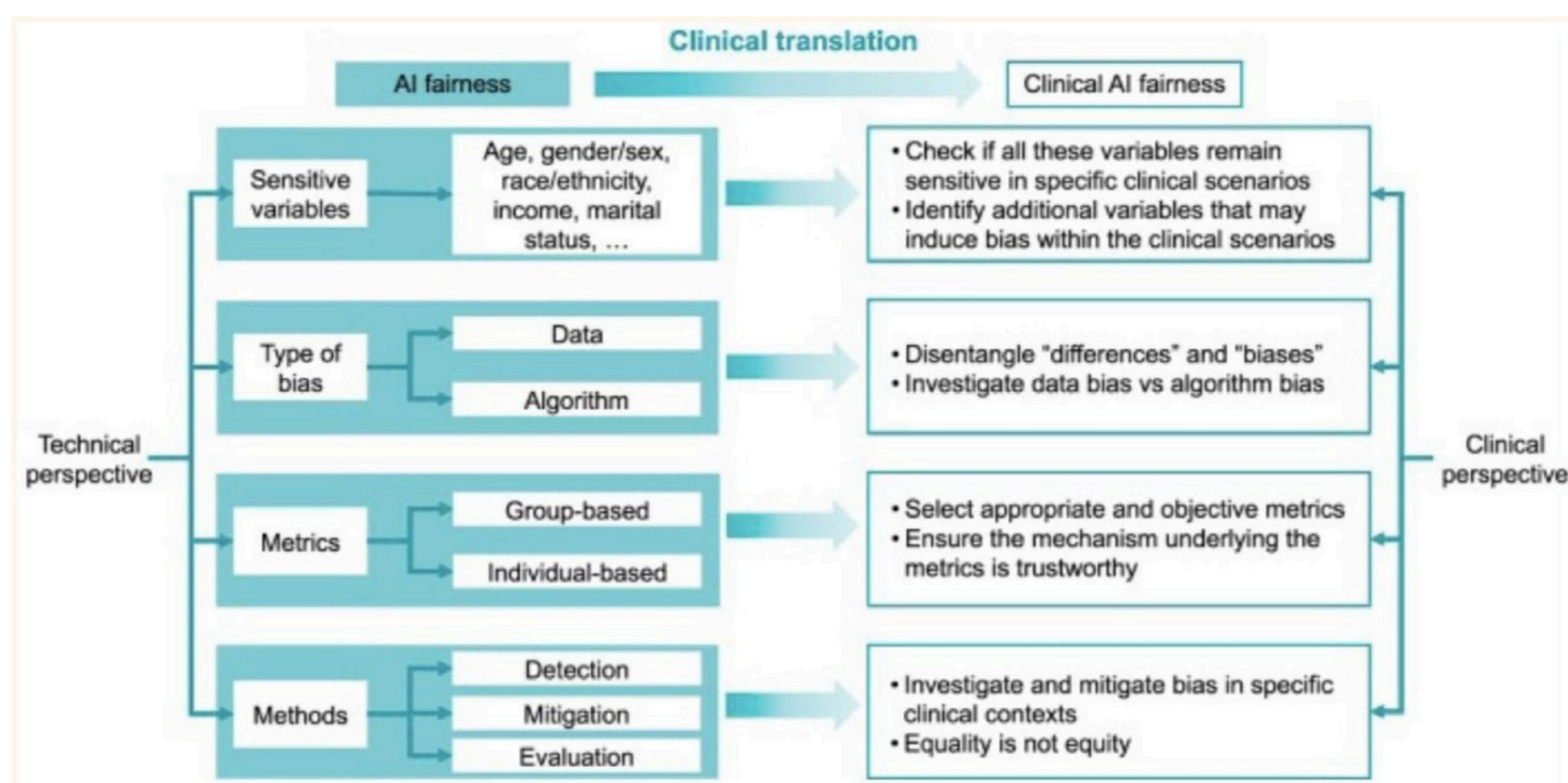
Develop frameworks for determining responsibilities and legal liability

Build infrastructure for safe data sharing

- As we consider some of these ethical challenges, there are different groups around the U.S. And certainly around the world that are working on developing potential best practices, but using a consensus approach. So trying to identify the very best practices that could benefit those globally.

AI FAIRNESS

- Conceptual model towards clinical AI fairness.



- Clinical fairness being a multidisciplinary research topic and requiring input from AI. Researchers, clinicians and ethicists.
- AI have expected to remain robust against cognitive biases and prejudices that humans experience of making judgments, and even to detect biases that humans cannot recognise.

Analyze Ethical and regulatory aspects



VULNERABLE POPULATIONS

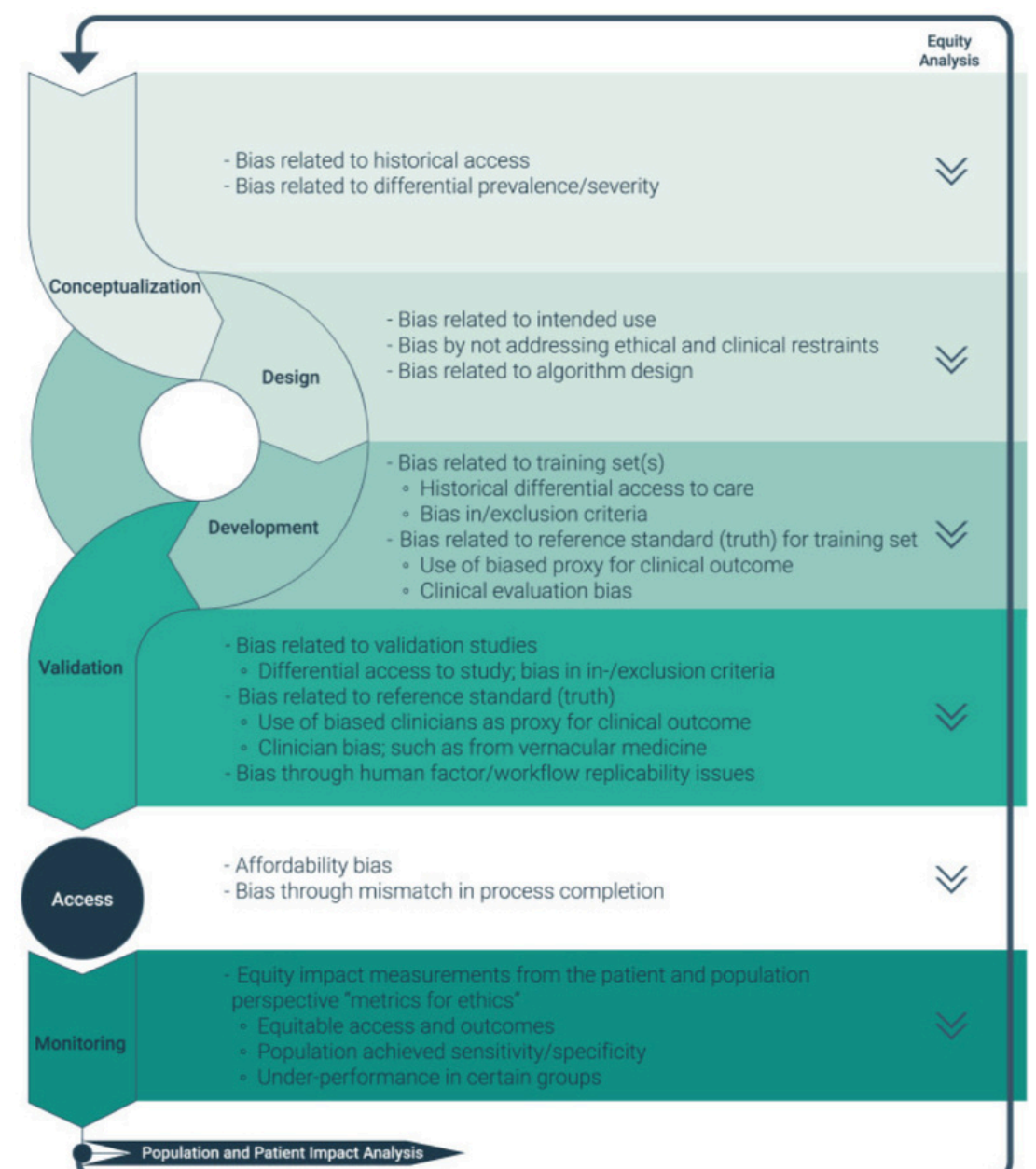
- ACCEPT-AI Framework: Key recommendations for pediatric data use in AI/ML research.

Age	• Include the patient's chronological age at the time of study enrollment • When applicable and available, include the patient's developmental age. If unavailable, state as such • Attempt to include developmental stages and relevant milestone metrics of CYP e.g., height and weight percentile upon enrollment, to capture the heterogeneity of participants. If developmental metrics are unavailable, state them as such • Include the age(s) of intended algorithm users e.g., pediatric only, pediatric and adult, or adult only
Communication	• Communication of study purpose to CYP as key stakeholders with developmentally appropriate communication strategies. • Communication with parents) or legal guardians as key stakeholders. • Tailor communication to social circumstance, addressing family complexities, including any court involvement. • Clear communication of technology-specific study purpose, risks, benefits, and alternatives with all key stakeholders. • Consider the use of videos, written material, and decision aids to facilitate education and enhance communication. • Involve stakeholders, including CYP and parents, in focus groups for design feedback where possible and relevant legal and institutional permissions are obtained. • State efforts taken to involve potential users in feedback of research idea and invest in community-level digital literacy. • Where possible, document and articulate model explainability.
Concept and assent	• Record mode of consent, who provided consent, e.g., parent, legal guardian, and how it was obtained. • Document any complex parental relations, dynamics, or court involvement that impact consent • Document children's social circumstances as relevant to safety, participation and evaluation. • For children in state-custody, ensure consent is obtained by relevant legal guardians or custodians and documented accurately. • Document relevant child protection laws pertinent to individual cases. • Attain assent when developmentally appropriate and/or required by regulations. • Ensure minors participate in the assent process in accordance with their developmental skills (e.g., appropriate modifications for children with clinically relevant developmental delay). • Record age when assent is provided. • Ensure local laws for adolescent assent/consent are followed.
Equity	• Ensure inclusion and exclusion criteria are clearly defined and specify disease, symptom, or condition of interest, with developmental stages considered as appropriate • State processes employed to reduce selection bias • Transparent demographic reporting, including race, documented sex, gender, and socioeconomic factors 9 • Provide details on how gender and documented sex have been incorporated into the study design 7 • Incorporate accessible research design to facilitate the inclusion of patients with disabilities (developmental and otherwise) • If skin tone could influence algorithmic outputs, ensure it is documented • Indicate the source of demographic information (e.g., self-reported) as well as details on non-reporting and missingness • Discuss the role of community engagement in the study
Protection of data	• State how data collection aligns with study objectives • State data-sharing plans when relevant • State if data is identifiable or de-identified • If data is de-identified, state compliance with the relevant legal frameworks, e.g., HIPAA, Common Rule, GDPR 24,25 • State data protection plans, addressing unique data risks in AI/ML including protections against cybersecurity breaches. • Disclose whether data can or cannot be retrieved/removed in the future by parents and CYP • Ensure social context of child e.g., suspected or confirmed child abuse or complex social circumstance is accounted for prior to any data releases that may involve parental requests or involvement, if available
TECHNOLOGICAL CONSIDERATIONS - (TRANSPARENCY OF TECHNIQUES TRAINING, AND TESTING METHODOLOGY)	• Ensure algorithmic studies are tailored to the needs of the pediatric population and clearly documented in the study protocol • Ensure AI/ML techniques are only used when potentially beneficial to the pediatric population, and that such benefits are clearly detailed in the study protocol • Detail any potential harms that pediatric subjects may incur as a result of the study • Identify measures taken to minimize risk to pediatric subjects throughout study and post-implementation • State measures taken to monitor and document adverse events that may affect pediatric subjects • State outcome measures and plans to clinically evaluate performance of algorithms on pediatric subjects • When available, utilize validated pediatric clinical scales in the clinical algorithm evaluation • Articulate how AI/ML will be trained to recognize/account for developmental heterogeneity • Document AI/ML methods using validated guidelines, e.g CONSORT-AI and SPIRIT-AI10,11 • Define data input and output (e.g., images, text) as well as the source (e.g., public dataset), and output • Account for age-specific factors related to disability and developmental conditions (e.g., natural disease progression) as relevant in study design, testing, and evaluation • State if the study involves adult, pediatric, or mixed data in training and/or testing • If the study involves both adult and pediatric data, state the purpose for this combination • If the study involves both adult and pediatric data, state whether the same or separate algorithms were used to assess each group



AI BIAS

- Bias can originate at various stages of the TPLC and can be addressed using specific metrics and tailored approaches.
- Following an expanded TPLC approach, equity analysis and bias mitigation will help stakeholders understand how bias may affect healthcare decisions and outcomes.
- Ai bias can originate at various stages of the total product lifecycle, and can be addressed using tailored approaches
- the goal is to initiate discussion involving various stakeholders, including bioethicists, A. I. Creators, regulatory agencies, patient advocacy groups, clinicians..
- What they think is that following an expanded total product lifecycle approach, equity analysis and bias mitigation can help stakeholders understand how bias might affect healthcare decisions and outcomes



Total Product LifeCycle (TPLC)
equity expanded framework
with examples for each phase.

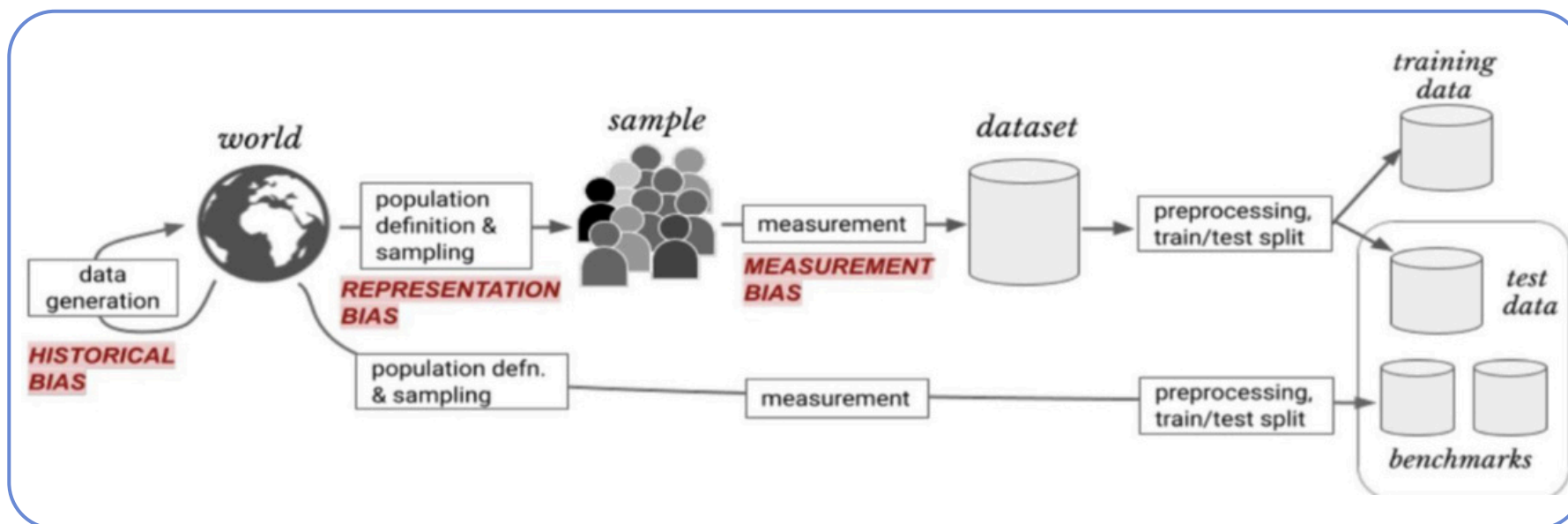
EXAMPLES OF AI BIAS SOURCES

- Biases in study inclusion criteria, e.g., using eGFR to select study participants
- Study data collection biases(including misclassification of race/ethnicity)
- Lack of representation / selection in the dataset
- Missing data
- Biases in imputed data
- Biases in learning and training data
- Collapsing race variables
- Lack of reporting for methodological approach

- Overfitting
- Lack of reporting for methodological approach
- Interpretation bias
- Correlation bias
- Training-validation data skew
- Lack of external validation
- Lack of performance assessments, such as calibration and discrimination
- Lack of reporting for methodological approach
- Insufficient sample size
- Labeling bias



DATA GENERATION

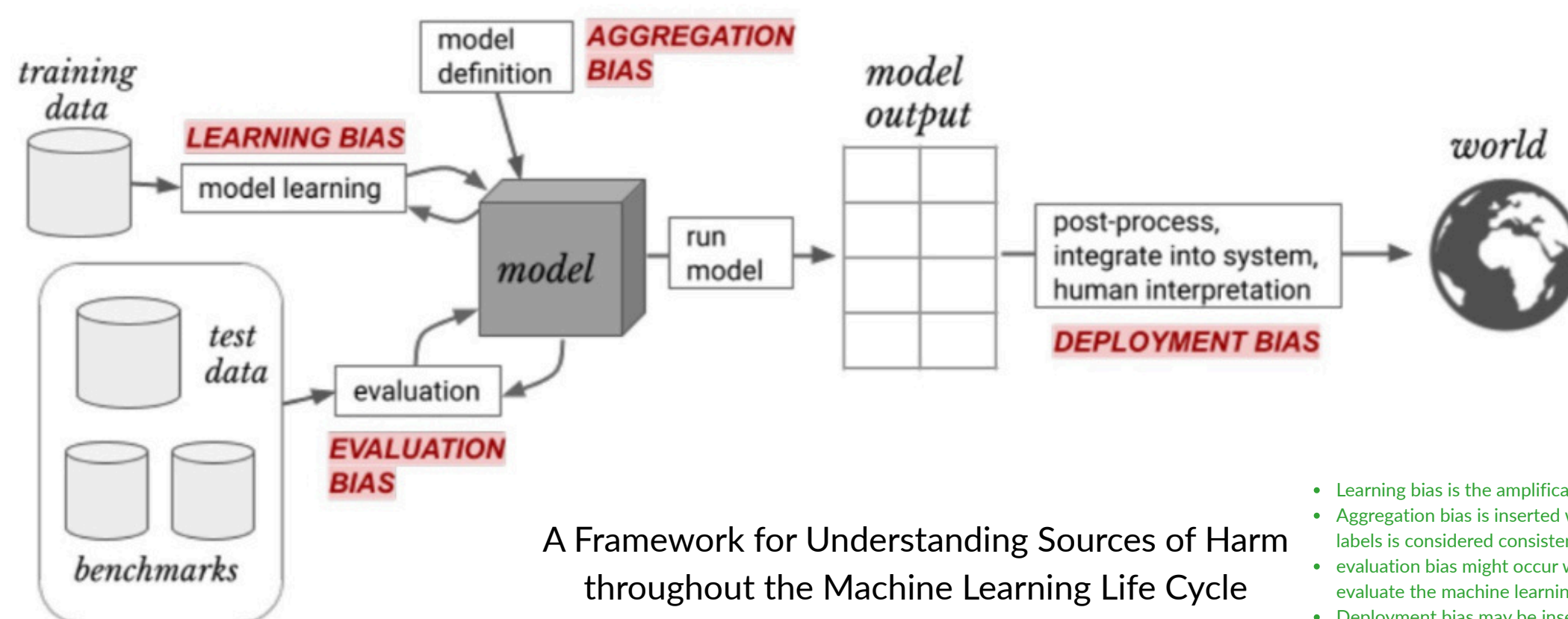


- Two examples of bias represented here appear during data generation.
- The historical or preexisting bias are existing societal stereotypes reflected in the data sets. Representation bias occurs when data are used are non-representative of the clinical populations leading to those systematic errors in machine learning model predictions and measurement bias surfaces where chosen features and labels are perfect proxies for the real variables of interest.

A Framework for Understanding Sources of Harm
throughout the Machine Learning Life Cycle
Harini Suresh & John Guttag. 2021.

Bias Type	Risk Description	Potential Mitigation	Resources
Historical/preexisting	Existing societal stereotypes reflected in datasets.	Debiasing Variational Autoencoder	<u>Amini et al., 2019</u>
Representation	Nonrepresentative data for population leading to systematic errors in ML model predictions	Targeted data augmentation	<u>Sontag & Johanssen, 2018</u>
Measurement/ cognitive	Chosen features and labels are imperfect proxies for the real variables of interest	Exchanging with domain experts about features	<u>Baer 2019</u>

MODEL BUILDING AND IMPLEMENTATION



A Framework for Understanding Sources of Harm
throughout the Machine Learning Life Cycle
Harini Suresh & John Guttag. 2021.

- Learning bias is the amplification of performance disparities .
- Aggregation bias is inserted with the incorrect assumption that mapping from inputs to labels is considered consistent across subsets of the data .
- evaluation bias might occur when the non representative testing population are used to evaluate the machine learning model .
- Deployment bias may be inserted when machine learning models are used in interpreted in different contexts on other than those that it was built for

MODEL BUILDING AND IMPLEMENTATION BIAS CONSIDERATIONS



Bias Type	Risk Description	Potential Mitigation	Resources
Learning	Amplification of performance disparities	Reevaluate model design choices	Hooker, 2021
Aggregation	Incorrect assumption that mapping from inputs to labels is consistent across subsets of the data	Coupled Learning Methods	Dwork et al., 2018
Evaluation	A non-representative testing population or inappropriate performance metrics are used to evaluate the ML model.	Representativeness of Benchmark Dataset Subgroup Validity	Giffen et al., 2021 Suresh & Guttag, 2019
Deployment	The ML model is used and interpreted in a different context than it was built for.	Holistic Evaluation Monitoring Plan & Human Supervision	Estiri et al., 2022 Giffen et al., 2022

AI EVALUATION: PROCESS STANDARDIZATION

Leveraging Clinical Research Pathway
Artificial Intelligence Translation in health care: Framework for
Evaluation and Documentation

Phase	Prepare: 0: Discovery and Intervention	Develop: 1. Technical Performance and Safety	Validate: 2. Efficacy and Side Effects	Deploy: 3. Therapeutic Efficacy	Maintain: 4. Safety and Effectiveness
AI Evaluation Priorities	Data Quality Evaluation User-centered design work and user requirements specification Evaluation of user requirements for sustainability and scalability	Model Development SaMD Review; IRB and FDA regulatory determination Model ethical and explainability assessment UX design research and prototype design Deployment Feasibility Analysis PCED submits the endorsed EHR Enhancement Request AI Ethics & Bias Evaluation	Iterative interface design with stakeholder focus groups Silent Evaluation/Pre-pilot model performance and validation Model implementation evaluation and validation Business plan validation and deployment planning	Prospective Study/Pragmatic Trial Clinical Validation Operational Education	Post-deployment surveillance and maintenance risk assessment Quality monitoring and audit Model maintenance plan Corrective and preventative action model documentation Fidelity/surveillance of clinical utility
AI Evaluation and Stewardship Checkpoint	Patient Impact Clinical Practice and Business Impact Societal Impact Purpose and Indications Model Planning and Architecture Historical Bias Evaluation Representation Bias Evaluation	Risk Assessment Model Accuracy Evaluation Usability Formative Measurement Bias Evaluation Learning Bias Evaluation	Silent Evaluation Model Statistical Validation Deployment and Validation Planning Aggregation Bias Evaluation	Clinical Validation User Education and Training Plan Monitoring and Reporting Plan Corrective and Preventative Action Plan Deployment Bias Evaluation	Post-Deployment Maintenance Quality Monitoring and Audit Maintenance De-implementation Continual Bias Evaluation Plan



The Role of Artificial Intelligence Model Documentation in Translational Science: Scoping Review

Tracey A Brereton ¹ ; Momin M Malik ¹ ; Mark Lifson ¹ ; Jason D Greenwood ² ; Kevin J Peterson ¹ ; Shauna M Overgaard ¹

Article	Authors	Cited by	Tweetations (3)	Metrics
Abstract - Introduction - Methods - Results - Discussion - References - Abbreviations - Copyright	Abstract Background: Despite the touted potential of artificial intelligence (AI) and machine learning (ML) to revolutionize health care, clinical decision support tools, herein referred to as medical modeling software (MMS), have yet to realize the anticipated benefits. One proposed obstacle is the acknowledged gaps in AI translation. These gaps stem partly from the fragmentation of processes and resources to support MMS transparent documentation. Consequently, the absence of transparent reporting hinders the provision of evidence to support the implementation of MMS in clinical practice, thereby serving as a substantial barrier to the successful translation of software from research settings to clinical practice.			

[Brereton et al., 2023](#)



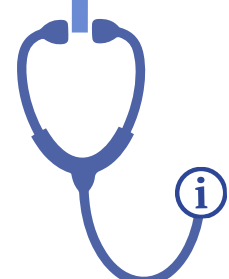
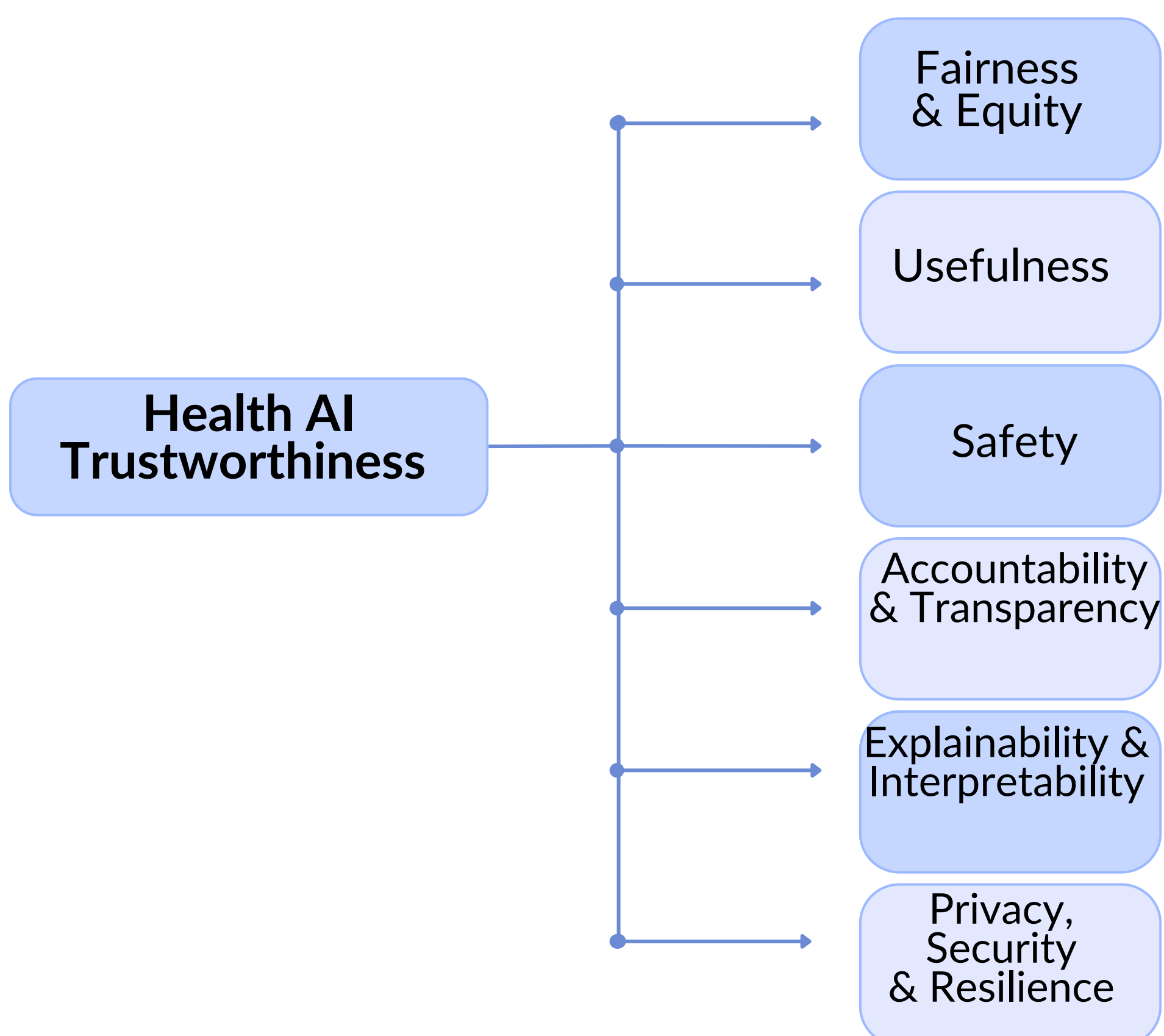


MODEL DOCUMENTATION FRAMEWORK

It is to work on expediting the translation of the AI solutions into clinical practice



CORE Fairness and Equity CHAI/NIST ALIGNED PRINCIPLES





- ## Work Groups



The Global Digital Health Partnership

- Artificial intelligence (AI)—the phenomenon of machines being able to solve problems that traditionally require human intelligence—has seen an enormous rise in interest due to considerable advances in effectiveness and use. The health sector, one of the most important sectors for societies and economies worldwide, is particularly interesting for AI applications, given the ongoing digitalization of health data and the promise for an improved quality of health and healthcare. However, due to the complexity of AI models, it is difficult to distinguish good from bad AI-based solutions and to understand their strengths and weaknesses. This is crucial for clarifying



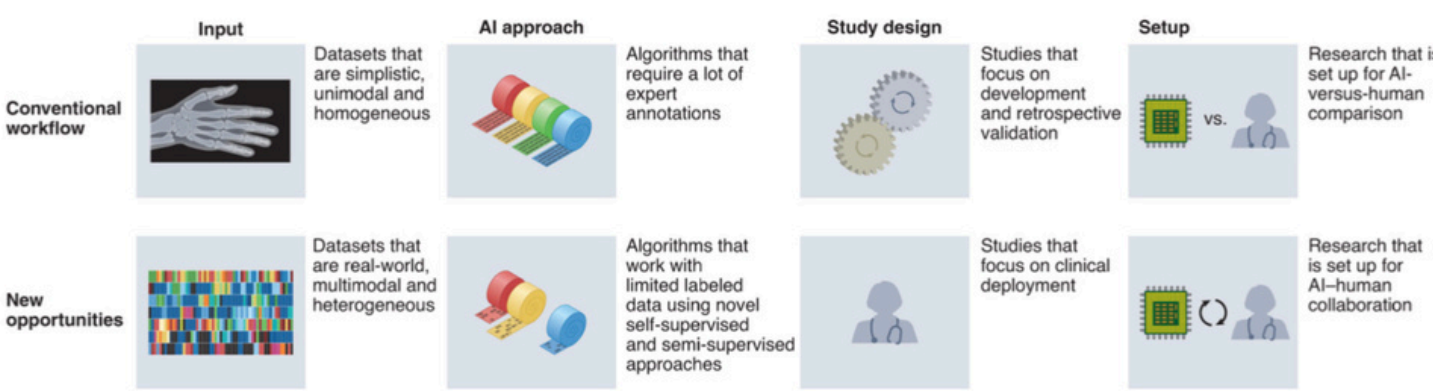
ANTICIPATE FUTURE AI TRENDS IN HEALTHCARE

FUTURE OPPORTUNITIES FOR AI IN HEALTHCARE



ADAPTIVE AI
GENERATIVE AI
DIVERSE USECASES

CONVENTIONAL VS. NEW



Opportunities for the development of AI algorithms.

Rajpurkar et al., 2022, Nat Med

PRINCIPLES FOR RESPONSIBLE DEVELOPMENT

1. Alleviate healthcare disparities	• What health disparities are reported for the present AI application? • How can the AI tool be designed to be accessible to and improve outcomes for the disadvantaged population? • What clinical interventions are needed to realize the benefit, and are these accessible? • How can data collection be supported in underserved communities for tool retraining over time?
2. Report clinically meaningful outcomes	• How is clinical benefit defined in this domain? • What is the present threshold for the clinical benefit of existing tools, and how can the AI tool improve upon this threshold?
3. Reduce overdiagnosis and overtreatment	• What disease state is an overdiagnosis? • For every case of overdiagnosis, what are the downstream costs to the patient and healthcare system? • How can this AI application reduce the number of overdiagnoses compared to existing approaches?
4. Have high healthcare value	• Is this AI tool addressing a high-priority healthcare need? • What would be the cost to the healthcare system in implementation, maintenance, and update? • What would be the cost to the patient who does and does not benefit from this tool? • Does this tool have high healthcare value, and if not, how can it be improved?
5. Incorporate biography	• What biographical data can be collected or carefully coded for the intended population? • How do these factors vary in the intended population? • How can these factors be included when developing AI tools?
6. Be easily tailored to the local population	• Can the training features be easily collected in different settings? • Are these features reliable for training across different populations? • Will the AI/ML workflow be made open-access?
7. Promote a learning healthcare system	• How will this AI application be evaluated over time, and at what intervals? • What are acceptable thresholds for performance? • How will the evaluation results contribute to continuous improvement?
8. Facilitate shared decision-making	• Have AI explainability tools been explored and utilized? • Do clinicians and patients find the explainability results helpful? • Have simpler, explainable algorithms been tried and compared to 'black-box' algorithms to determine if a simpler model performs just as well? • How can patient values be easily integrated into the use of the AI tool?

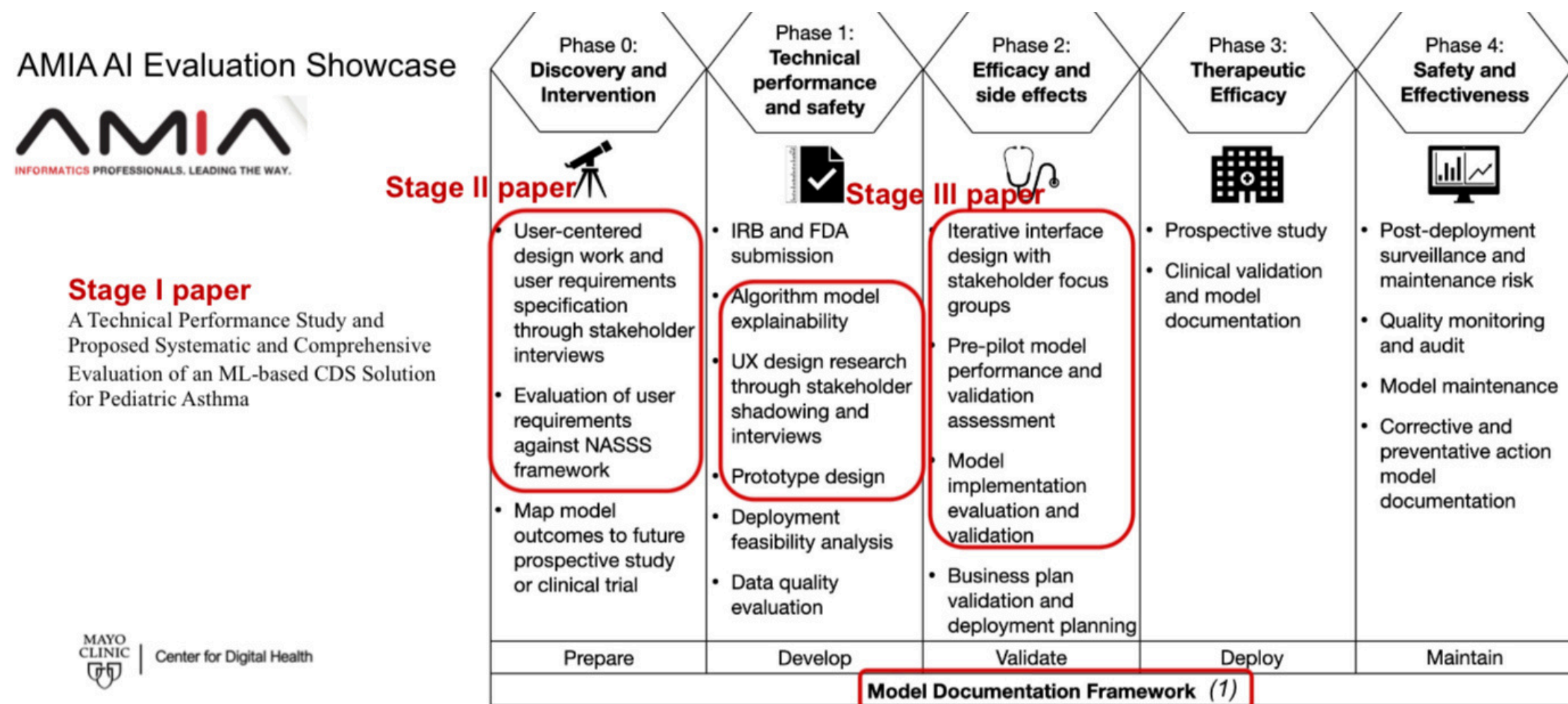
Questions that can be used when considering each principle in the AI development process.

Badal et al., 2023, Commun Med



ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

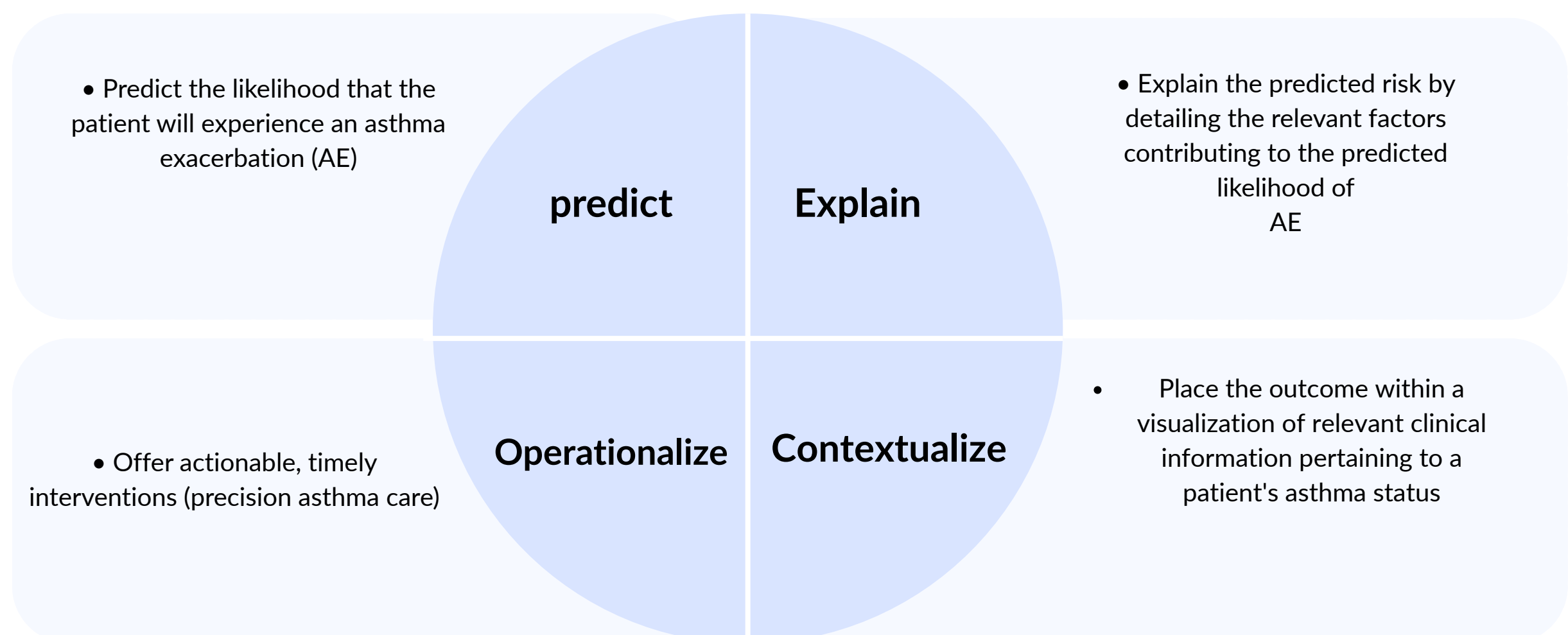
PHASED RESEARCH FRAMEWORK OF A-GPS



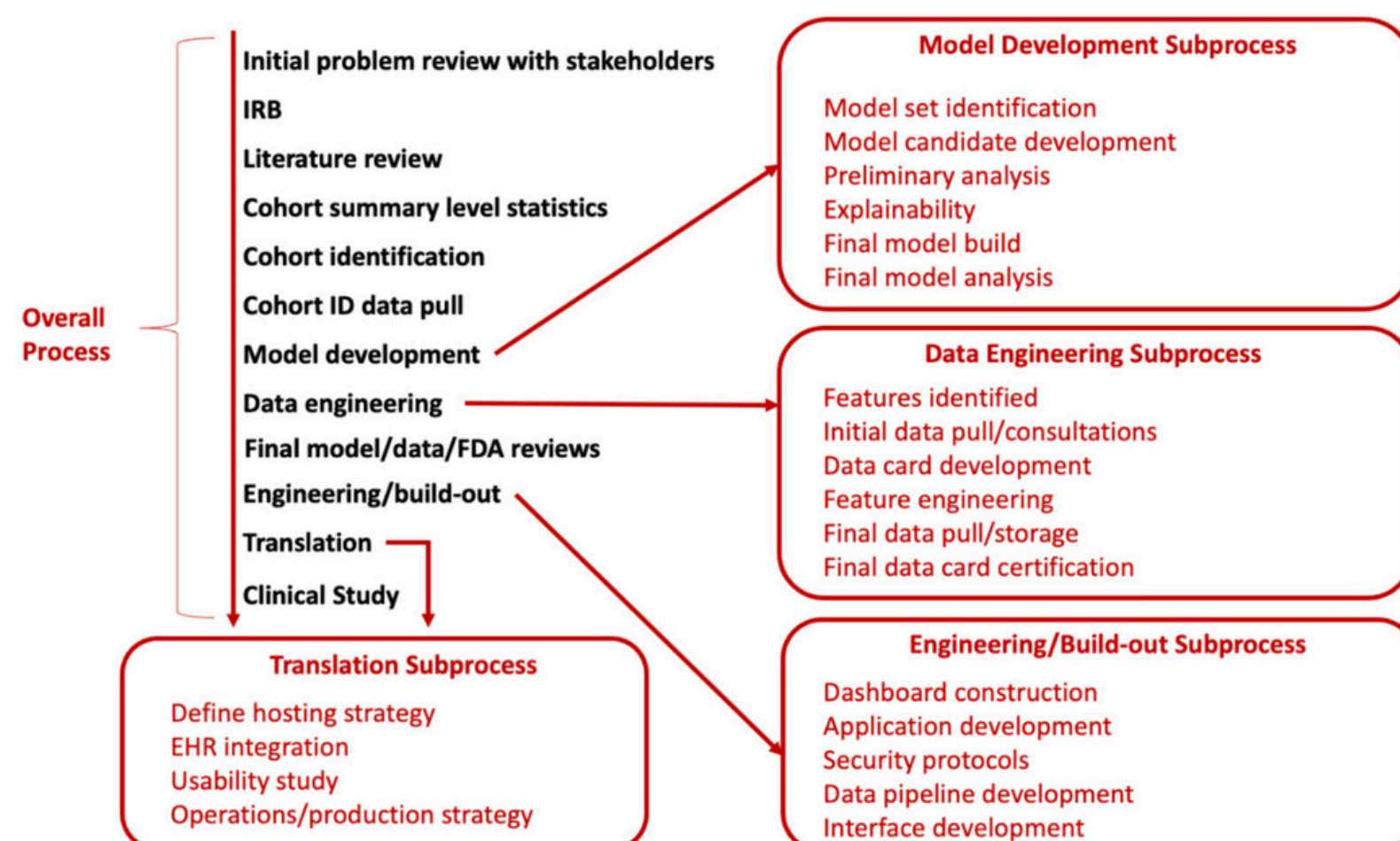
overgaard et al., 2022

Rajpurkar et al., 2022, Nat Med

AIM OF A-GPS (ASTHMA GUIDANCE AND PREDICTION SYSTEM)



Methodology



overgaard et al., 2022



ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

Model Development

cohort Evaluation

Patients aged 6-17, dx active asthma (clinical visit with an associated asthma diagnosis in last 12 months ≥ 1)



Model Feature Selection

Identified as clinically significant factors by the domain experts

28 variables which include both structured and unstructured data

Unstructured: NLP, MedTagger IEarchitecture

Structured: Queried Mayo Database using SQL



Data Quality Evaluation

Data definition and category for each feature

Data provenance and distribution of data availability(missingness)

Distribution of comorbidities aligned with clinical criteria, clinical description, and interpretation.



Model Evaluation

Conducted using 5 fold cross-validation

Binary dependent variable evaluation: area under the receiver operating characteristic curve (AUC-ROC)

F1 evaluation of output, 0.5 positive/negative threshold

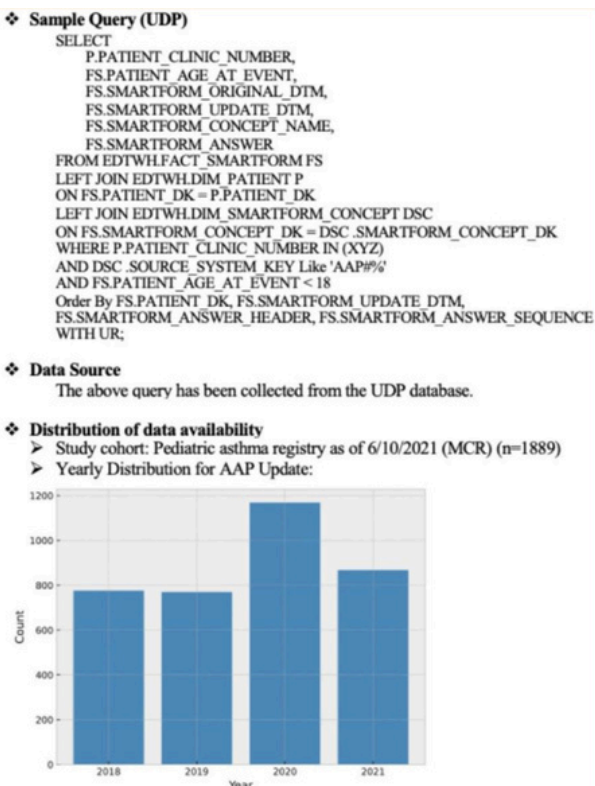
overgaard et al., 2022

Metadata Prototype

Metadata Prototype			
Overall capabilities: FAIR principles, metadata completeness reports, traceability, usage metrics, version control, data governance, comment section for each card aspect, suggested metadata elements based on other projects, executable, searchable and discoverable, actionable and reproducible, bottom-up crowd sourcing			
	Project Card	Data Card	Model Card
Scope	The Project Card contains the project description, inclusion/exclusion criteria for project scope, and includes suggestions for inclusion/exclusion criteria from an electronic phenotype database (PhEB). The Project Card sets the objective and scope of the project, describing the overall clinical intent, limitations, clinical departments involved, etc.	The Data Card contains a feature description, distributions over time, frequencies within the study cohort and population subsets, associated code sets, data availability and completeness, and data source. The Data Cards will be model-agnostic in order for Data Cards to be reusable across projects.	The Model Card includes the AI model description, inputs and outputs of the model, model architecture, limitations, trade-offs, missingness handling technique, and model performance. The model is dependent on the data features used in the model, therefore there are many Data Cards to one Model Card. The Model Card operates on the defined population.
	clinicaltrials.gov	Mayo Clinic Data Card Template	Model Cards for Model Reporting
	Design description: The Clinical Trials website was created to "provide patients, family members, health care professionals, researchers, and the public easy access to information on publicly and privately supported clinical studies." Each study is maintained and updated appropriately to ensure relevancy and transparently communicate the description of a clinical trial. The NIH and FDA collaborated to create this registry to "test the effectiveness of experimental drugs for serious or life-threatening diseases or condition" after the FDAMA. The components included in each study follows a defined set of standard reporting protocols that compiles a summary about the study. Design components: Study Description Study Design Groups and Cohorts Outcome Measures	Template description: A Data Card represents example metadata that would be needed for documenting the features of a machine learning model. Data Cards support transparency, accountability, and explainability the reporting of model features by standardizing the documentation of feature descriptions. Data Card components: Project Information Structured Data (Feature Name...) Unstructured Data (Feature Name...)	Journal description: Model Cards are a documentation framework accompanying ML models that acts as a transparency tool for model reporting by providing descriptions of functionality and performance metrics while also including metrics that capture bias, fairness and inclusion considerations. Model Cards aim to provide a concise, holistic view of ML models by explaining what it does, its intended audience, who maintains it, model construction and architecture, and information on the training data, detailing a model's performance metrics as well as limitations and risk mitigations. Model Card components: Model Details Intended Use Factors and Metrics Evaluation Data

overgaard et al., 2022

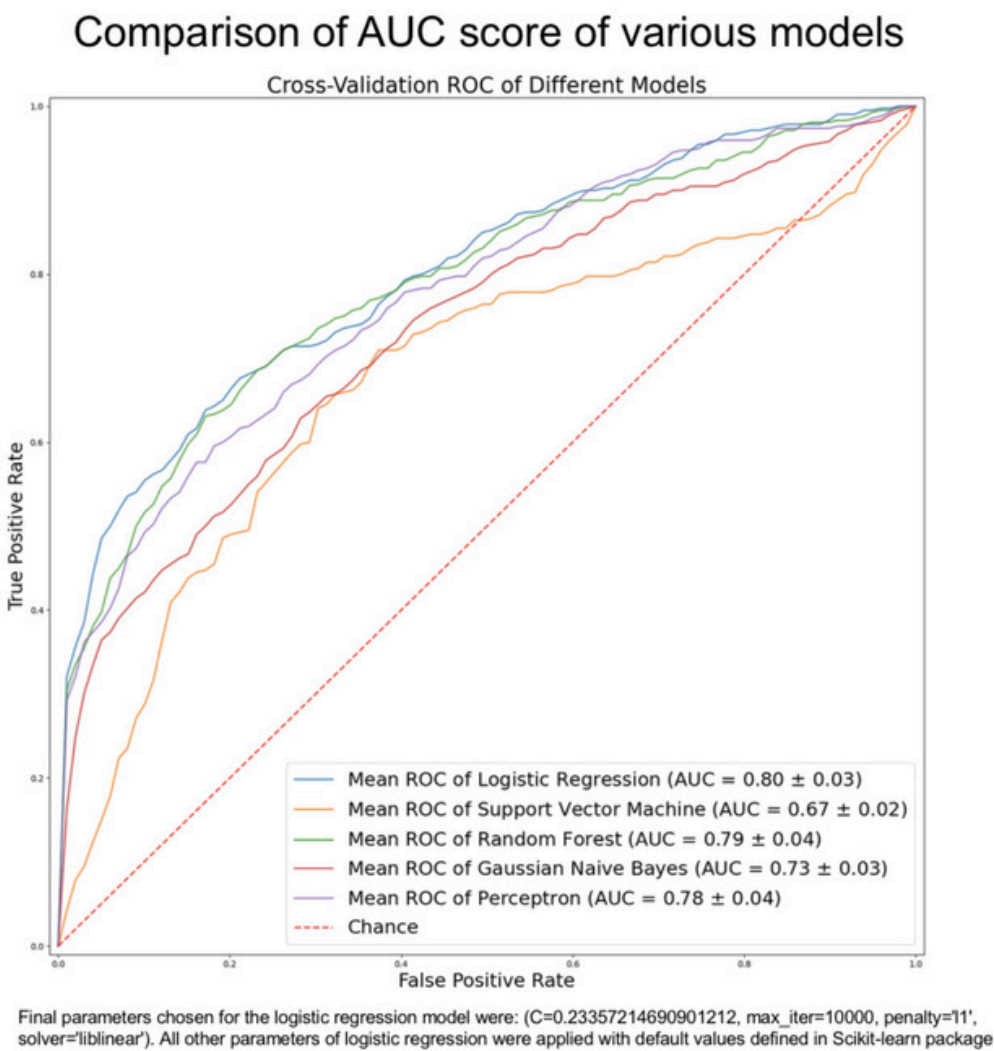
Data Card Example





ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

Results



Prediction Task

Predict asthma exacerbation (AE) within the next 12 months

Logistic Regression model output

Threshold	Mean F1
0.1	0.421527
0.2	0.522789
0.3	0.531575
0.4	0.484955
0.5	0.456748
0.6	0.430991
0.7	0.4161
0.8	0.388956
0.9	0.339465

MULTIDISCIPLINARY TEAM

- United objective to provide solutions that are:
 1. People centric
 2. Value added
 3. Evidence-based

Multidisciplinary Team to Facilitate Translation				
Translational Informatics Analysts	Clinicians	Data science	Ethicists	MLOps
Profuct Owner	Operational Staff	Patients	Quality Management Team	UX Researchers and designers

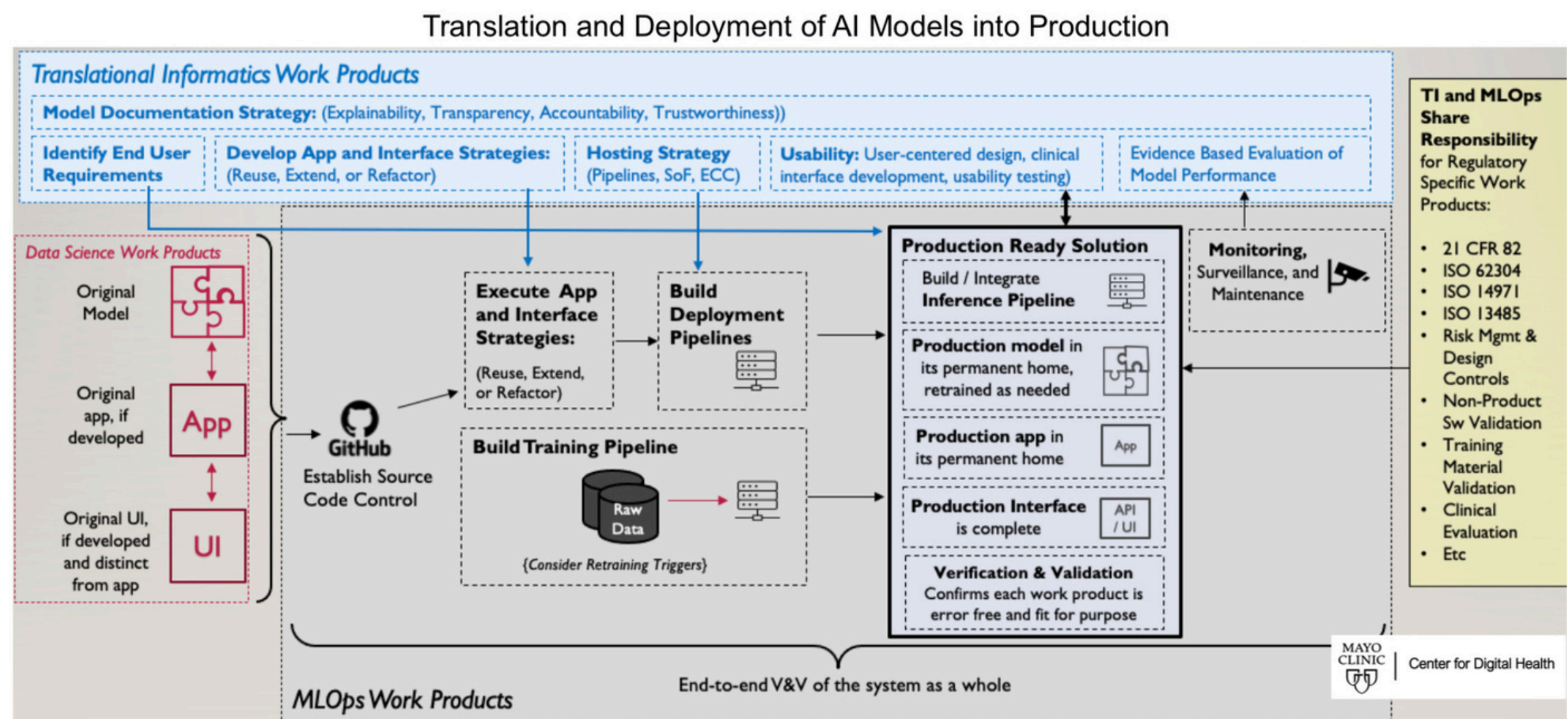
[overgaard et al., 2022](#)





ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

TRANSLATION AND DEPLOYMENT OF A-GPS



A Proposed Model Documentation Framework to Facilitate Translation of AI Models in Healthcare

Tracey A. Brereton¹, Shauna M. Overgaard, Ph.D.¹, Kristelle A. Jose¹, Chung II. Wi, M.D.^{2,3}, Kevin J. Peterson, Ph.D., M.S.¹, Amin Nikakhtar, Ph.D., M.S.¹, Young J. Juhn, M.D.^{2,3}

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What might the attendee be able to do after being in your session?

Attendees will be equipped to apply a Model Documentation Checklist to guide AI/ML (Artificial Intelligence / Machine Learning)-based clinical decision support (CDS) tool documentation, encompassing multidisciplinary best practice reporting requirements to facilitate the translation of explainable and transparent AI/ML tools in healthcare. The Model Documentation Checklist is a collaborative tool that a) follows phases of product lifecycle and AI clinical evaluation, b) serves to iteratively develop a comprehensive AI model documentation deliverable, and c) rigorously addresses translation, implementation, and accountability gaps in oversight and standardization by leveraging reporting guidelines and published best practices.^{1,2,3-6}

Description of the Problem or Gap

Literature surrounding the current state of AI/ML-based CDS documentation recognizes inconsistencies in standards and governance that do not effectively balance CDS tool intelligence with explainability, therefore creating gaps in translation, implementation, and accountability.² CDS tools integrated within clinical settings show evidence of accompanying documentation with levels of rigor and transparency regarded as insufficient by field experts and justifications involving fairness, long-term reliability, and usefulness as limited, leading to biased, untrustworthy, black-box tools.^{7,8} The lack of sufficient documentation and inability to leverage available resources to communicate functionality proves challenging for CDS tools to be explainable from technological, legal, medical, and patient perspectives and must be enhanced to effectively translate tools into clinical settings.⁹

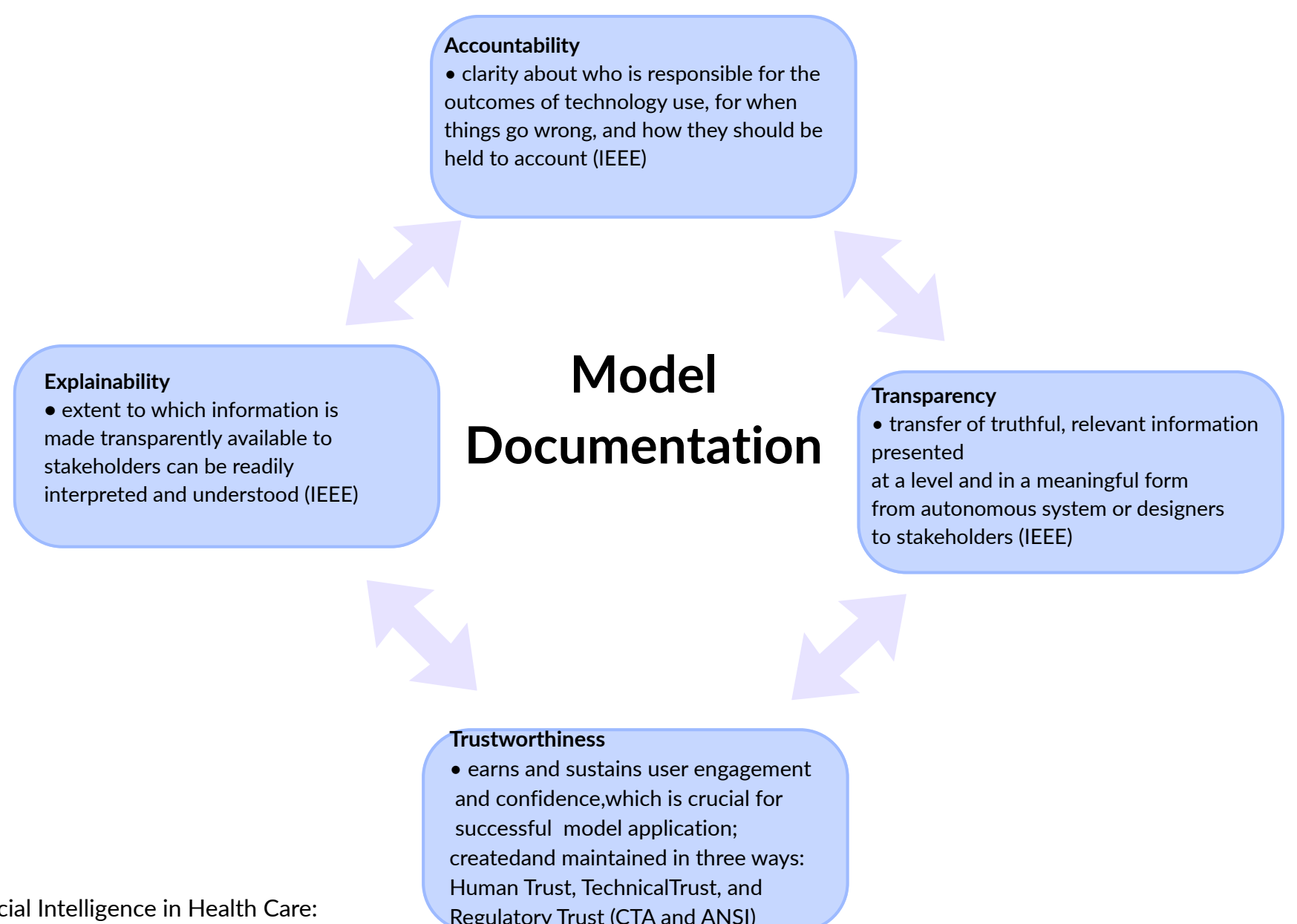
Methods

To satisfy a real need in our organization's translational AI work, our interdisciplinary team sought to create a comprehensive model documentation framework, from which we derived an accompanying Model Documentation

Brereton et al., 2022

AI/ML Documentation

- primary values driving model documentation framework: explainability, transparency, accountability, trustworthiness
- Harmonizes AI development and ethical considerations
 - Complements and works in tandem with model development
 - Communicates model components and features
 - Addresses limitations and risks
 - Avoids complex "black-box" models
- Value and document transparency and accountability in model development to standardize explainable AI and gain trust





ANALYZE CASE STUDIES IN REAL-WORLD EXAMPLES

Model Documentation framework

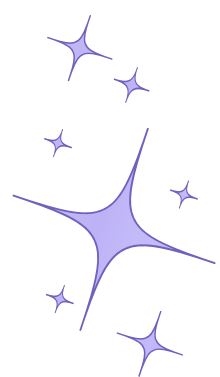
Model Documentation Checklist Items - Phase 1 - Prepare						
Patient Impact						
Completed / Reviewed	Assigned To	Index	Checklist Item / Description	Reporting Guideline	Section of Reporting Guideline	Referenced It
		1-1	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models	TRIPOD	Background and Objectives	3a
		1-2	Specify the objectives, including whether the study describes the development or validation of the model or both	TRIPOD	Background and Objectives	3b
		1-3	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centers	TRIPOD	Participants (Methods)	5a
		1-4	Describe eligibility criteria for participants	TRIPOD	Participants (Methods)	5b
		1-5	Give details of treatments received, if relevant	TRIPOD	Participants (Methods)	5c
		1-6	Explain the intended use of the AI intervention in the context of the clinical pathway, including its purpose and its intended users (e.g. healthcare professionals, patients, public).	CONSORT-AI	Background and Objectives	2a
		1-7	Specify objectives, endpoints, and hypotheses	CONSORT-AI	Background and Objectives	2b
		1-8	Provide examples of out-of-scope use cases and note limitations of generalizability	Model Card	Intended Use	

Tool Link

Model Documentation framework

Phase 1 (prepare)	Phase 2 (Develop)	Phase 3 (Validate)	Phase 4 (Deploy)
Phase subgroups: patient impact , purpose , indications	Phase subgroups: model planning and architecture , risk assessment , usability formative	Phase subgroups: deployment planning , clinical validation	Phase subgroups: User education and training , monitoring and reporting , corrective and preventative Actions





SUMMARY



AI: it is computer system that recognize patterns

AI	Stimulation of human intelligence and beyond using programmatic routines
ML	Subset of AI uses statisical algorithm to represent complex patterns within data
DL	Subset of ML uses layered artificial neural networks capable of emergently representing latent properties of the data

Why use AI in healthcare

Improve patient outcomes	Automate processes and save time	Reduce Costs
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AI key concepts

self supervised learning	learning from unlabeled data by leveraging information extracted from data itself
semi supervised learning	learning from a small amount of labeled data combined with a large amount of unlabeled data
causal inference	finding the effect of a component or treatment on a system using data
Reinforcement learning	learning in an interactive environment using feedback from actions and past experiences

Overview of the progress , challenges and opportunities for AI in health care

Progress	Deployment progress: (CMS reimbursement , FDA approvals , Prospective trials) Medical image interpretation: (Radiology , Pathology , Ophthalmology , Gastroenterology)
Challenges	Fairness: (Ethical data use , Equity and bias) accountability: (Regulations , Responsibilities) Implementation challenges: (Model trust , Data limitations)
Opportunities	Beyond Supervised learning (weakly supervised , semi supervised , unsupervised , image reconstruction) Nonimage data source (multimodal , molecules , signals , electronic health care)





1-Which of the following is a key limitation of machine learning?

A. It cannot automate processes

B. It cannot learn patterns

C. It cannot replace clinical intuition and qualitative assessment

2-Which of the following is described as a subset of AI that uses statistical algorithms to represent complex patterns and allow a computer to learn from data?

A. Deep Learning

B. Machine Learning

C. Artificial Intelligence

3-In the "SHARED ACCOUNTABILITY" model, which group is responsible for vendor assessment and integration of an AI tool?

A. Physician

B. Developer

C. Healthcare Org.

4-Which of the following is a key principle for the responsible development of AI in healthcare?

A.Maximizing profits for AI companies

B.Reducing overdiagnosis and overtreatment

C.Eliminating the need for all human clinicians

5-In the asthma AI model development example, which step involved checking the completeness, consistency, and clinical alignment of the data?

A.Cohort Evaluation

B.Model Feature Selection

C. Data Quality Evaluation

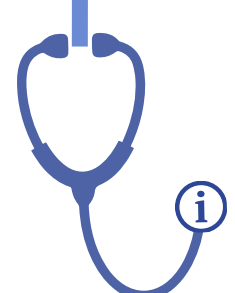
6-In the Model Documentation Framework, which phase specifically focuses on Deployment Planning and Clinical Validation?

A.Phase 2: Develop

B.Phase 3: Validate

c.Phase 4: Deploy

1-C. 2-B. 3-C. 4-B 5-C 6-B





Team Leaders

Shahad Alshehri



Khalid Alghamdi

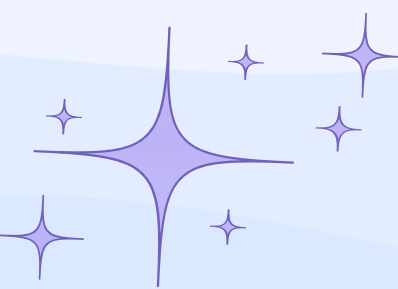


Team Members

Seeta bin aqeel



Abdulelah Aljadaan



سبحان الله، الحمد لله، لا إله إلا الله، الله أكبر،
لا حول ولا قوة إلا بالله

